



RIPHAH
INTERNATIONAL UNIVERSITY

Department of Software Engineering

Software Requirements Specification

Hospital Management System

Version 1.0

#	Name	SAP ID
1	Muhammad Habib	62669
2	Mamoon Ishaq Khan	62338
3	Daniyal Karim	63780

Course: Software Requirements Engineering

Instructor: Sir Muhammad Shabir Hassan

Semester & Section : SE-4-1

1. Introduction

1.1 Purpose

This document is the Software Requirements Specification for the Hospital Management System. It describes the functional and non functional requirements, system architecture, use cases, and diagrams for the proposed system. This document serves as the foundation for design, development, and testing of the HMS.

1.2 Scope

The Hospital Management System is a web based software solution that digitizes and connects all core hospital operations into one secure and centralized platform. The system covers patient registration, appointment scheduling, clinical care, laboratory services, pharmacy management, billing, and system administration

The following are outside the scope of this document: actual software development and coding, hardware setup and configuration, mobile application development, and external insurance system integrations.

1.3 Definitions and Acronyms

Term	Definition
HMS	Hospital Management System
SRS	Software Requirements Specification
UC	Use Case
OPD	Out Patient Department
IPD	In Patient Department
ER	Entity Relationship
DFD	Data Flow Diagram
RBAC	Role Based Access Control
PDF	Portable Document Format
LAN	Local Area Network

1.4 Document Overview

- Section 1 introduces the project, its purpose, scope, and key definitions.
- Section 2 describes the stakeholders, actors, existing system problems, and proposed solution.
- Section 3 lists all functional and non-functional requirements.

- Section 4 describes the system architecture and all system diagrams.
- Section 5 presents all use case diagrams, activity diagrams, and use case description tables organized by module.

1.5 References

- Riphah International University Software Requirements Engineering Course Material
- Hospital Management System Project Proposal Group Document

1.6 Elicitation Techniques

To understand what the system should do our team visited two hospitals in Islamabad and talked to the staff working there. We prepared a set of questions and asked them to the staff at each hospital.

1.6.1 Field Visits

We visited the following hospitals

1. Ayesha Foundation Hospital,
2. Haris Family Hospital,

We went there in person to see how they currently work and to ask the staff about their daily tasks and problems.

1.6.2 Questionnaire Based Interviews

We prepared a list of questions and asked them to different staff members including receptionists, doctors, pharmacy staff, billing staff, and lab technicians. The questions were about how they register patients, book appointments, handle billing, manage medicines, and process lab tests.

1.6.3 What We Found

After visiting both hospitals we found that

- Everything was done on paper which was slow and caused mistakes
- Receptionists had trouble finding old patient records quickly
- Billing was done manually and often had errors
- Doctors could not see a patients past history easily
- Pharmacy stock was not connected to doctor prescriptions

These problems helped us decide what features the system must have.

1.6.4 Why We Used This Method

We chose questionnaires and field visits because they are simple and effective. Hospital staff are busy during working hours, so we could not do long interviews. Visiting in person also helped us see the actual problems that staff face every day.

RIPHAH INTERNATIONAL UNIVERSITY

2. System Overview

2.1 Stakeholders

These are the stakeholders those interact with the Hospital Management System

Stakeholder	Role in the System
Patient	Registers in the system, books appointments, and views bills and lab reports
Doctor	Accesses patient records, records diagnoses, writes prescriptions, and requests lab tests
Receptionist	Manages patient registration, appointment scheduling, and ward management
Admin	Manages users, doctors, departments, and staff
Pharmacy Staff	Tracks medicine stock and fulfills digital prescriptions
Billing Staff	Generates invoices automatically and tracks payments
Lab Technician	Processes lab test requests and uploads results
Administrator	Manages system level tasks such as backups and health checks

2.2 Actors

No	Actor	Description
1	Patient	Registers in the system, books appointments, and views own lab reports and bills
2	Receptionist	Registers patients, manages appointments, admits and discharges patients
3	Doctor	Views patient records, diagnoses patients, prescribes medicines, and requests lab tests
4	Lab Technician	Processes lab test requests and uploads test results
5	Pharmacist	Verifies prescriptions, dispenses medicines, and manages inventory
6	Accountant	Generates bills, records payments, issues receipts, and generates financial reports
7	Admin	Manages users, doctors, departments, and staffing levels

8	Administrator	Manages system level operations such as backup and health monitoring
9	System	External actor that sends automated notifications and verifies credentials

2.3 Existing System and Current Situation

Most hospitals currently rely on handwritten records, physical folders, and basic spreadsheets to manage daily operations. While this approach may work at a very small scale it creates serious problems as the number of patients and staff grows.

- Patient data is stored in physical folders that are difficult to find quickly especially during emergencies
- Manual appointment scheduling frequently results in double bookings and scheduling conflicts
- Billing is calculated by hand leading to errors, delays, and patient dissatisfaction
- Doctors cannot access a patient medical history instantly during consultations or emergencies
- Pharmacy stock is tracked separately and is not linked to prescriptions causing dispensing errors
- Management has no real time visibility into hospital operations, finances, or staff performance

2.4 Problem Statement

How can hospitals replace slow and error prone paper based workflows with a single easy to use software system that connects patient records, appointments, billing, pharmacy, and clinical care while reducing administrative effort and improving the quality of patient care?

2.5 Proposed Solution

We propose the Hospital Management System, a centralized web based platform that integrates all core hospital functions into one secure system. Every department works within the same platform and shares data in real time, eliminating the need for paper records and disconnected tools. The system is organized into ten core modules covering patient registration, appointment scheduling, ward management, clinical care, laboratory services, pharmacy, billing, financial reporting, system administration, and a patient self service portal.

2.6 Assumptions and Constraints

Assumptions like as

- All hospital staff have basic computer literacy and can use a web browser
- The hospital has a stable internet connection and a local area network
- All patient data entered into the system is accurate and verified by the responsible staff member
- The system will be hosted on a secure server managed by the hospital IT team

Constraints are

- The system is a web based application and does not include a mobile application
- The system does not integrate with external insurance systems
- The laboratory module does not include specialized diagnostic equipment integration
- This SRS document covers requirements only and does not include actual software development

3. System Requirements

3.1 Functional Requirements

Functional requirements describe what the system must do. The following requirements define all the actions and behaviors the HMS must support

FR No	Requirement
FR-01	The system shall allow all staff members to log in using a unique username and password
FR-02	The system shall verify user credentials and grant access based on the assigned role
FR-03	The system shall allow staff members to reset their password through a registered email address
FR-04	The system shall allow the admin to create, edit, and delete user accounts
FR-05	The system shall allow the admin to assign roles and permissions to each user
FR-06	The system shall allow the admin to manage doctor profiles including specialization and department
FR-07	The system shall allow the admin to create and manage hospital departments
FR-08	The system shall allow the receptionist to register new patients with personal and medical details
FR-09	The system shall assign a unique ID to each registered patient
FR-10	The system shall allow the receptionist to update existing patient information
FR-11	The system shall allow the receptionist or patient to book, cancel, and reschedule appointments
FR-12	The system shall check for scheduling conflicts before confirming an appointment
FR-13	The system shall allow the receptionist to add patients to a waitlist when no slots are available
FR-14	The system shall allow the receptionist to admit patients and assign them to a ward and bed
FR-15	The system shall allow the receptionist to transfer patients between wards
FR-16	The system shall allow the doctor or receptionist to discharge a patient after all bills are cleared
FR-17	The system shall allow the doctor to view complete patient history and medical records

FR-18	The system shall allow the doctor to record diagnoses and clinical observations
FR-19	The system shall allow the doctor to create digital prescriptions for patients
FR-20	The system shall allow the doctor to request lab tests and specify test parameters
FR-21	The system shall allow the lab technician to process lab test requests and upload results
FR-22	The system shall notify the doctor and patient when a lab report is uploaded
FR-23	The system shall allow the pharmacist to verify prescriptions before dispensing medicines
FR-24	The system shall allow the pharmacist to issue medicines and update inventory automatically
FR-25	The system shall allow the pharmacist to set restock alerts for medicines with low stock
FR-26	The system shall allow the pharmacist to track and flag medicines nearing their expiry date
FR-27	The system shall allow the accountant to generate itemized bills for patient services
FR-28	The system shall allow the accountant to apply insurance coverage to reduce patient bills
FR-29	The system shall allow the accountant to record payments and issue receipts
FR-30	The system shall allow the accountant and admin to generate financial reports
FR-31	The system shall allow the administrator to perform database backups manually or automatically
FR-32	The system shall allow the administrator to monitor system health and performance
FR-33	The system shall automatically send notifications for appointments, lab results, and low stock alerts
FR-34	The system shall allow patients to view their own profile, lab reports, and bills through the portal

3.2 Non Functional Requirements

Non functional requirements describe how well the system must perform. The following requirements define the quality standards the HMS must meet

NFR No	Category	Requirement
--------	----------	-------------

NFR-01	Performance	The system shall support at least 500 concurrent users without any slowdown or degradation in performance
NFR-02	Security	The system shall enforce role-based access control so each user can only access what their role permits
NFR-03	Security	The system shall encrypt all passwords and sensitive patient data stored in the database
NFR-04	Speed	The system shall respond to all common user actions in under 3 seconds under normal load
NFR-05	Usability	The system shall be easy enough for non technical hospital staff to use without requiring special training
NFR-06	Reliability	The system shall maintain an uptime of at least 99 percent as hospitals operate 24 hours a day 7 days a week
NFR-07	Scalability	The system shall be designed to handle growing patient and transaction data over multiple years without requiring major changes
NFR-08	Data Backup	The system shall perform automatic daily backups of all data and allow manual backups at any time
NFR-09	Compatibility	The system shall work correctly on all major web browsers including Chrome, Firefox, Edge, and Safari
NFR-10	Availability	The system shall be accessible 24 hours a day and 7 days a week with minimal planned downtime
NFR-11	Maintainability	The system shall be built with a modular design so individual modules can be updated or replaced without affecting the rest of the system
NFR-12	Data Integrity	The system shall prevent duplicate patient records and ensure all data entered is validated before being saved

4. System Architecture and Diagrams

4.1 System Architecture

The Hospital Management System follows a three tier client server architecture. This design separates the system into three layers and makes the system easier to maintain, scale, and secure.

Layer	Description
Presentation Layer (Frontend)	This is what the user sees and interacts with. It includes all web pages and forms used by staff and patients
Application Layer (Backend)	This layer processes all user requests, applies business rules, and coordinates data between the frontend and the database
Data Layer (Database)	This layer stores all system data including patient records, appointments, prescriptions, bills, and lab reports

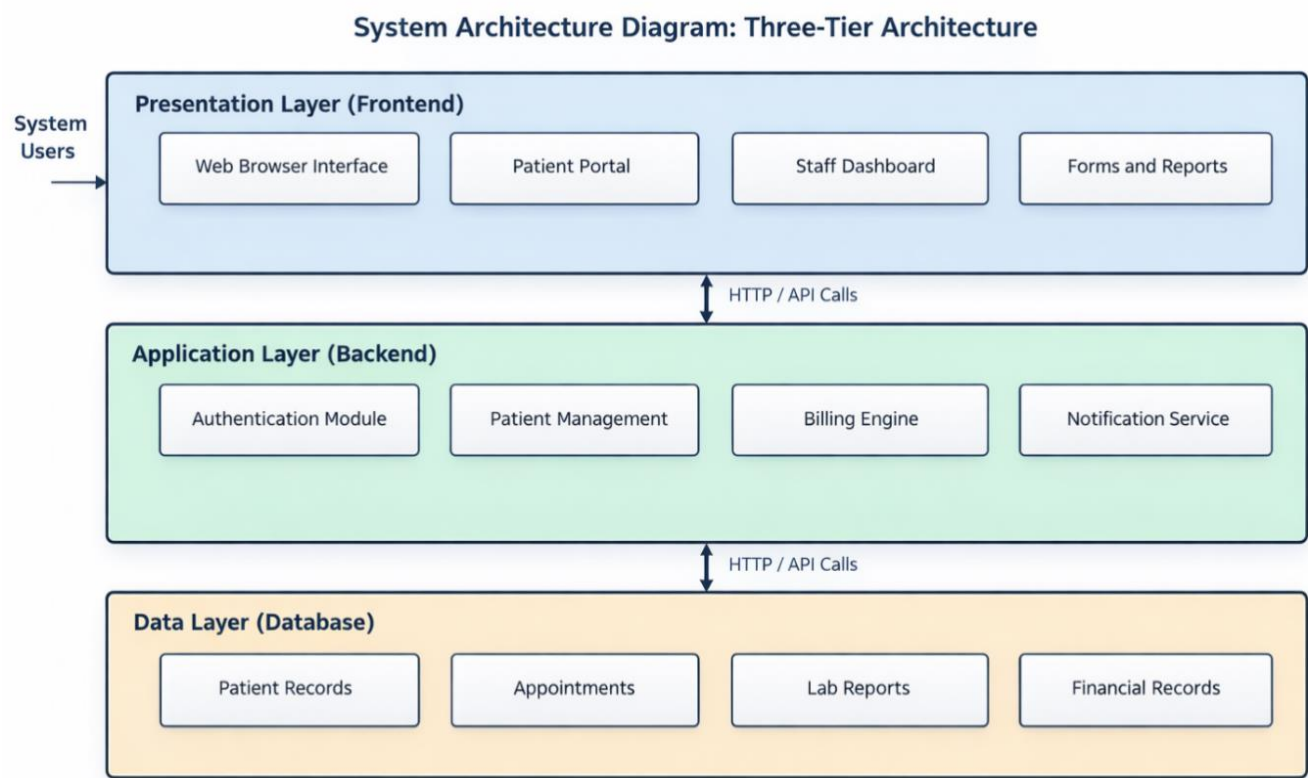


Figure 1: System Architecture Diagram

4.2 External Interface Requirements

4.2.1 Hardware Interfaces

- The system requires a standard personal computer or laptop with at least 4 GB RAM
- A stable network connection is required for all users to access the web based system
- A printer is required for generating and printing bills, prescriptions, and reports

4.2.2 Software Interfaces

- The system runs on standard web browsers including Chrome, Firefox, Edge, and Safari
- The system requires a web server to host the application
- The system uses a relational database to store and manage all records

4.2.3 Communication Interfaces

- The system communicates over a standard Local Area Network or Wide Area Network
- All data transmission between the client browser and the server must be encrypted using HTTPS

4.3 Data Flow Diagram Level 0

The Level 0 DFD shows the Hospital Management System as a single process and illustrates the flow of data between the system and its external actors.

Data Flow Diagram: Level 0 (Context Diagram)

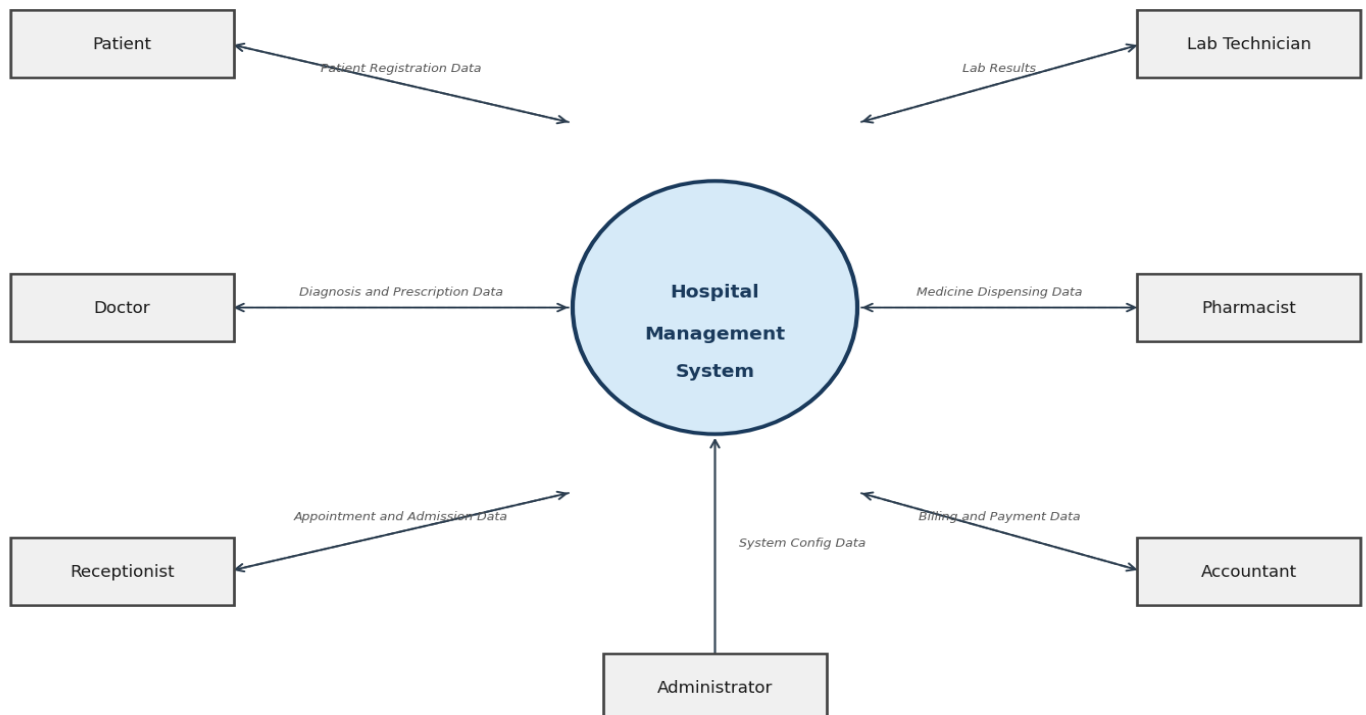


Figure 2: Data Flow Diagram Level 0 (Context Diagram)

4.4 Data Flow Diagram Level 1

The Level 1 DFD breaks down the main system process into its core sub processes and shows how data flows between each module internally.

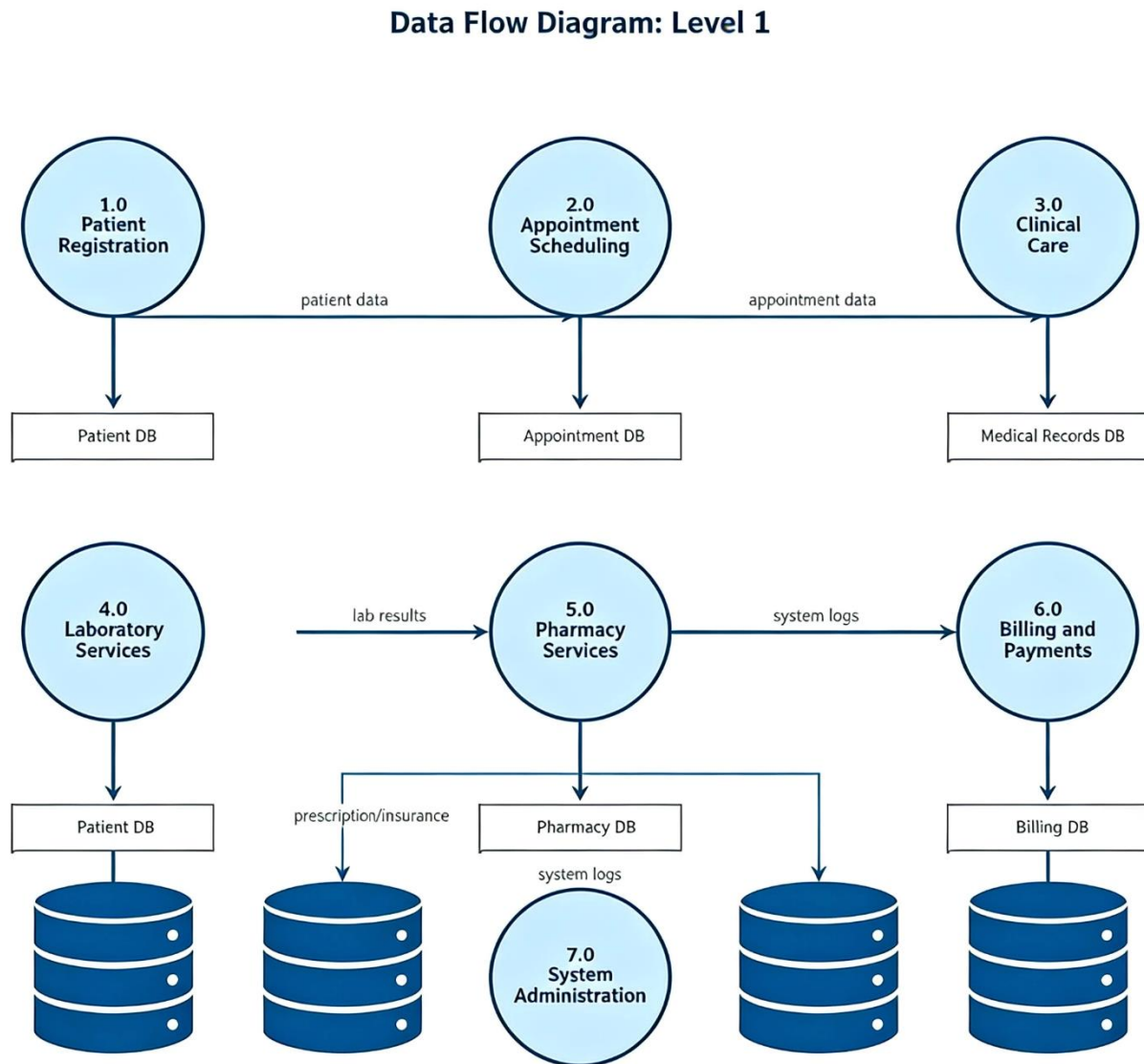


Figure 3: Data Flow Diagram Level 1

4.5 Entity Relationship Diagram

The ER Diagram shows the main data entities in the system and the relationships between them.

Entity Relationship Diagram: Hospital Management System

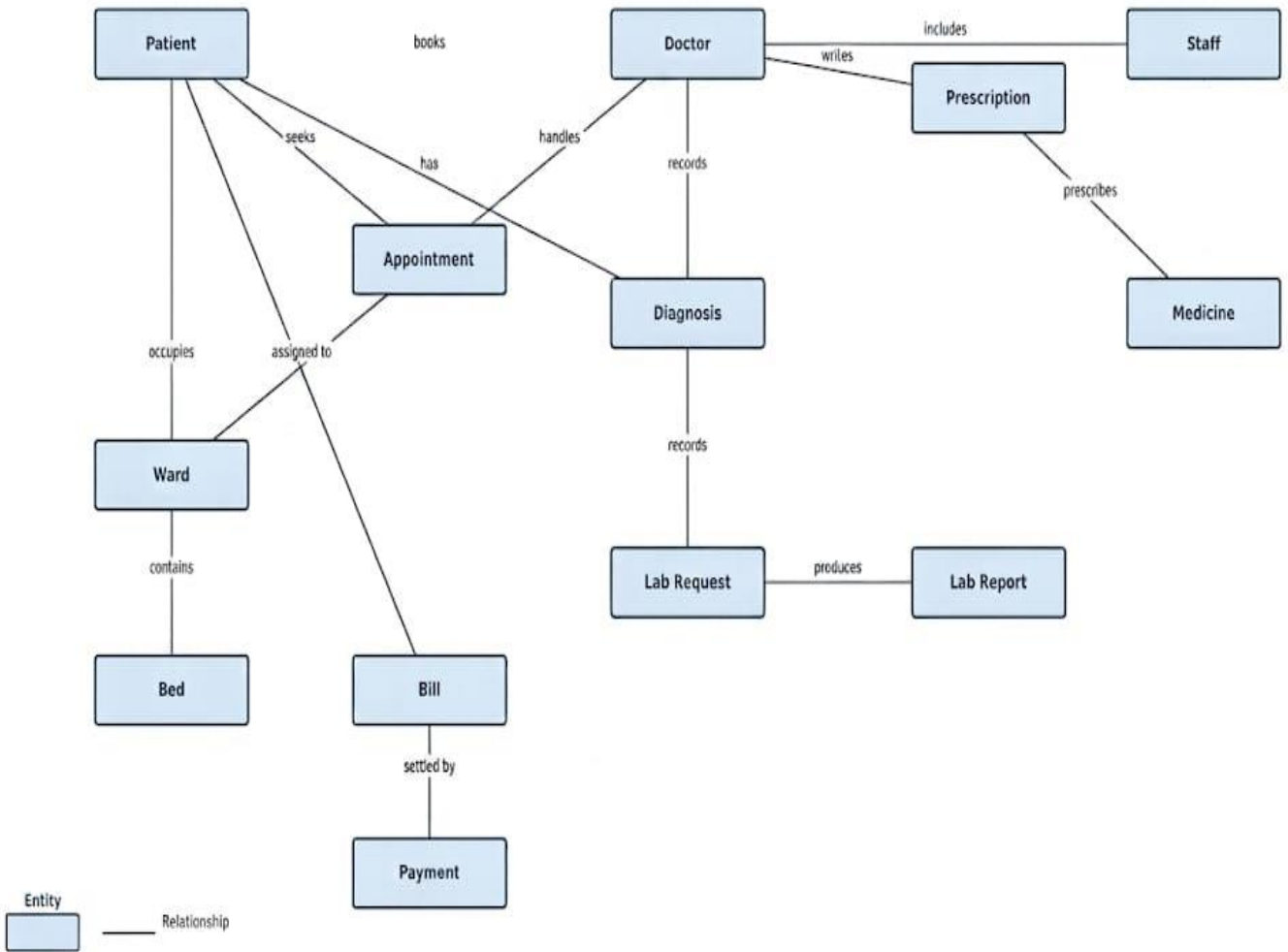


Figure 4: Entity Relationship Diagram

5. Use Case Specifications

This section presents the use case diagrams, activity diagrams, and use case description tables for all modules of the Hospital Management System.

5.1 Overall Use Case Diagram

The following diagram shows all actors and all modules of the Hospital Management System

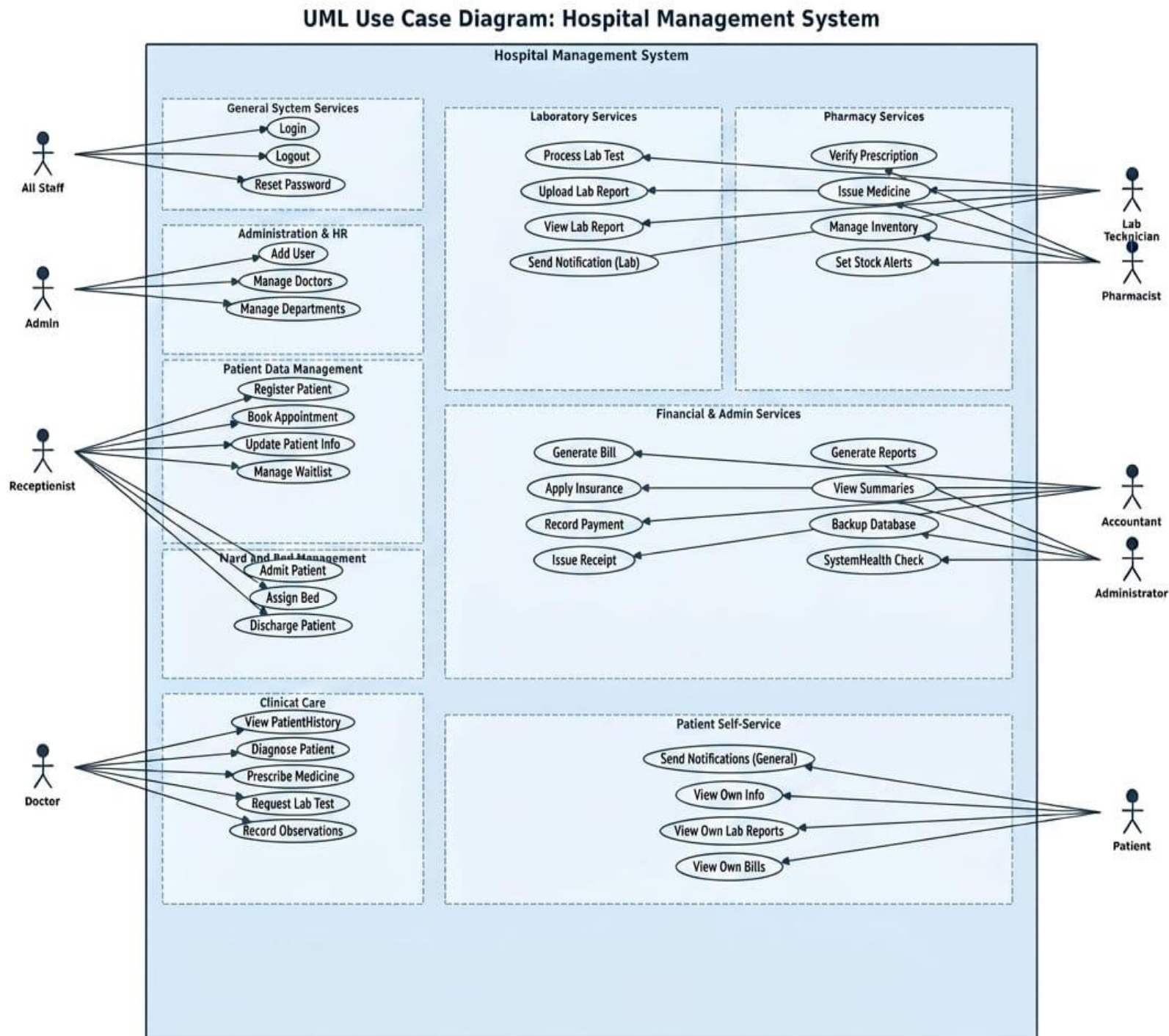


Figure 5: Overall Use Case Diagram

5.2 General System Services

Use Case Diagram

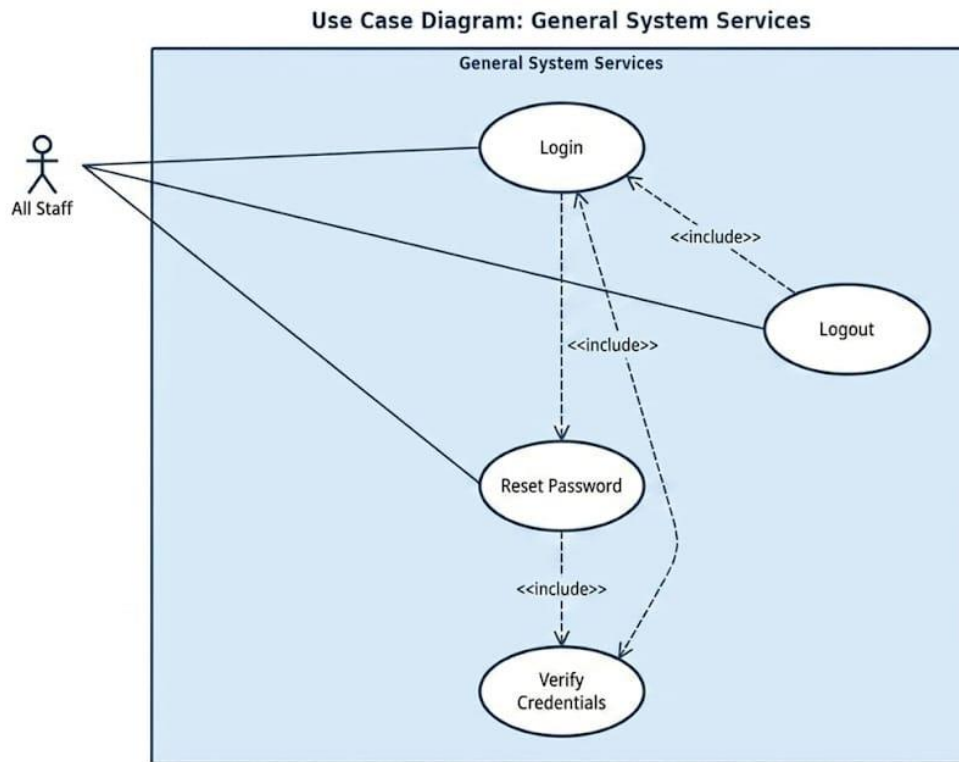


Figure 6: General System Services Use Case Diagram

Activity Diagram

Activity Diagram: General System Services

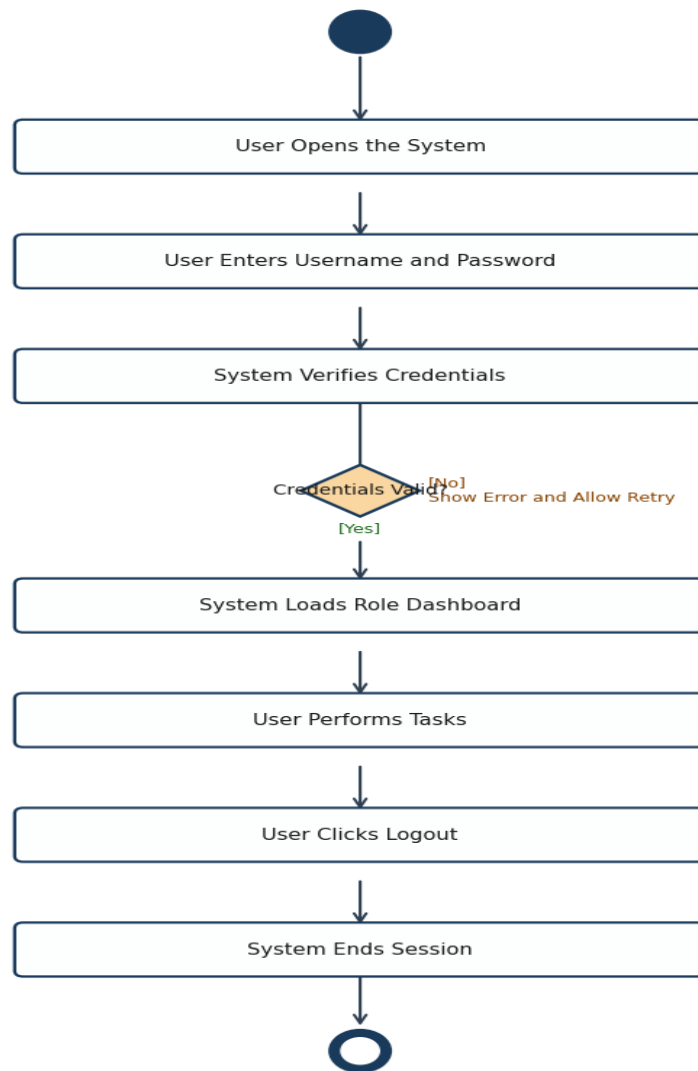


Figure 7: General System Services Activity Diagram

Use Case Description Tables

UC-01 Login

Use Case ID	UC-01
Name	Login

Actors	All Staff
Priority	High
Trigger	User opens the system and clicks Login
Summary	Allows all staff members to access the system using their username and password
Pre-Conditions	User must have a registered account in the system
Post-Conditions	User is successfully logged in and redirected to their role-based dashboard
Special Requirements	Password must be encrypted. Session expires after 10 minutes of inactivity
Basic Flow	
Actor Action	System Response
1. User enters username and password	2. System verifies the entered credentials
3. User clicks the Login button	4. System grants access and loads the dashboard
5. User starts using the system	6. System records the login activity in the audit log
Alternative Flow	
1-A. User enters wrong password	2-A. System shows an error message and allows retry. After 3 failed attempts the account is temporarily locked

UC-02 Logout

Use Case ID	UC-02
Name	Logout
Actors	All Staff
Priority	High
Trigger	User clicks the Logout button
Summary	Allows all staff members to safely end their session and exit the system
Pre-Conditions	User must be logged in

Post-Conditions	User session is ended and user is redirected to the login page
Special Requirements	All unsaved data must be auto saved before logout
Basic Flow	
Actor Action	System Response
1. User clicks the Logout button	2. System ends the current session
3. User is redirected to the login page	4. System clears all session data
Alternative Flow	
1-A. User closes the browser without logging out	2-A. System automatically ends the session after 10 minutes of inactivity

UC-03 Verify Credentials

Use Case ID	UC-03
Name	Verify Credentials
Actors	System
Priority	High
Trigger	User submits login credentials
Summary	System automatically checks if the entered username and password are correct
Pre-Conditions	User must have entered a username and password
Post-Conditions	System confirms whether credentials are valid or invalid
Special Requirements	Credentials must be checked against encrypted records in the database
Basic Flow	
Actor Action	System Response
1. System receives the entered credentials	2. System searches the database for matching records

3. System compares entered password with stored encrypted password	4. System returns a valid or invalid result
Alternative Flow	
1-A. No matching username found	2-A. System returns an invalid credentials message

UC-04 Reset Password

Use Case ID	UC-04
Name	Reset Password
Actors	All Staff
Priority	Medium
Trigger	User clicks Forgot Password on the login page
Summary	Allows staff members to reset their password if they have forgotten it
Pre-Conditions	User must have a registered account with a valid email address
Post-Conditions	User password is successfully reset and the user can log in with the new password
Special Requirements	Password reset link must expire within 24 hours
Basic Flow	
Actor Action	System Response
1. User clicks Forgot Password	2. System displays the password reset form
3. User enters their registered email address	4. System sends a password reset link to the email
5. User clicks the reset link and enters a new password	6. System updates the password and confirms the change
Alternative Flow	
1-A. User enters an email that is not registered	2-A. System shows a message that the email was not found

5.3 Administration and HR

Use Case Diagram

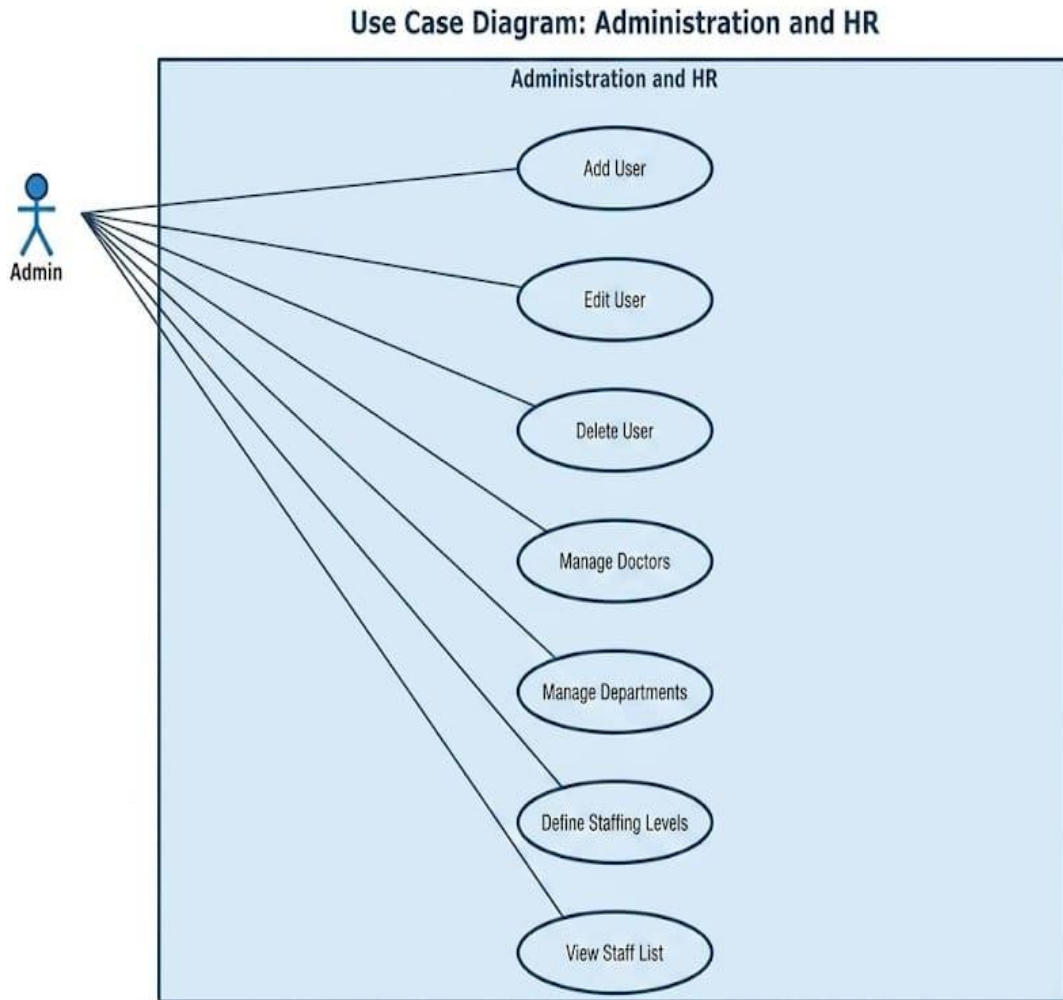


Figure 8: Administration and HR Use Case Diagram

Activity Diagram

Activity Diagram: Administration and HR

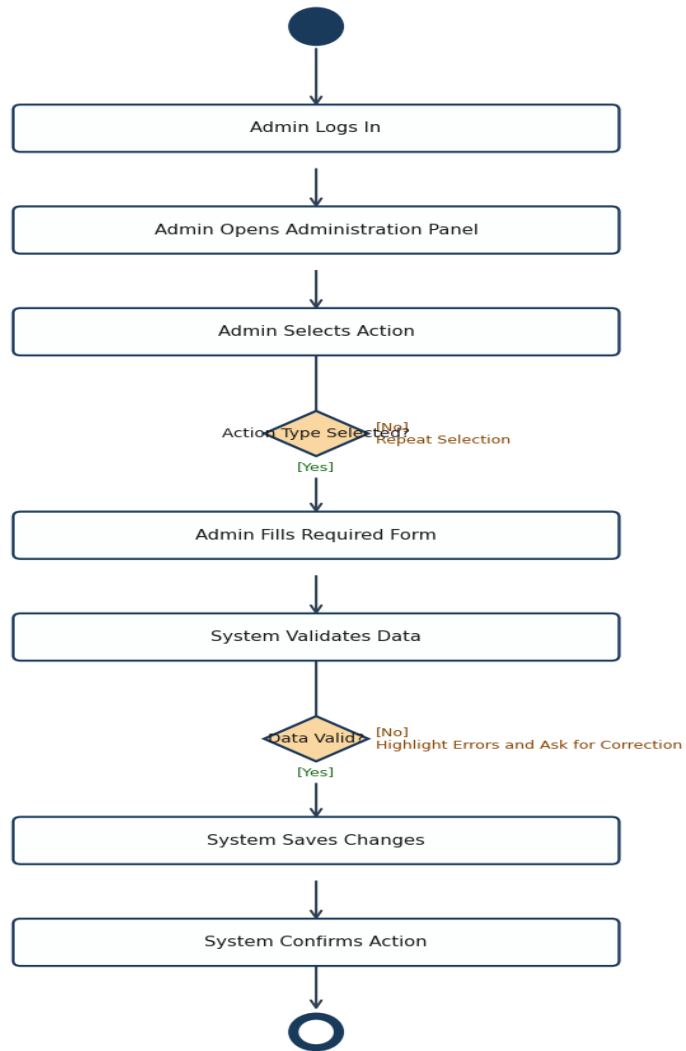


Figure 9: Administration and HR Activity Diagram

Use Case Description Tables

UC-05 Add User

Use Case ID	UC-05
Name	Add User
Actors	Admin
Priority	High

Trigger	Admin clicks Add New User in the user management section
Summary	Allows the admin to create a new user account and assign a role in the system
Pre-Conditions	Admin must be logged in with admin privileges
Post-Conditions	New user account is created and the user can log in with their assigned credentials
Special Requirements	Each user must have a unique username and a role assigned
Basic Flow	
Actor Action	System Response
1. Admin clicks Add New User	2. System displays the user creation form
3. Admin enters user details and assigns a role	4. System validates the entered information
5. Admin submits the form	6. System creates the account and sends login credentials to the user
Alternative Flow	
1-A. Admin enters a username that already exists	2-A. System shows a duplicate username error and asks for a different username

UC-06 Edit User

Use Case ID	UC-06
Name	Edit User
Actors	Admin
Priority	Medium
Trigger	Admin selects a user and clicks Edit
Summary	Allows the admin to update an existing user account details or role
Pre-Conditions	Admin must be logged in. The user account must already exist
Post-Conditions	User account is updated with the new information

Special Requirements	Changes to user role take effect on the next login
Basic Flow	
Actor Action	System Response
1. Admin searches for and selects a user	2. System displays the current user details
3. Admin updates the required fields	4. System validates the updated information
5. Admin saves the changes	6. System updates the user record and confirms the change
Alternative Flow	
1-A. Admin enters invalid data	2-A. System highlights the invalid fields and asks for correction

UC-07 Delete User

Use Case ID	UC-07
Name	Delete User
Actors	Admin
Priority	Medium
Trigger	Admin selects a user and clicks Delete
Summary	Allows the admin to remove a user account from the system
Pre-Conditions	Admin must be logged in. The user account must exist
Post-Conditions	User account is permanently removed and the user can no longer log in
Special Requirements	Deleted user data should be archived before removal for audit purposes
Basic Flow	
Actor Action	System Response
1. Admin searches for and selects a user	2. System displays the user details with a Delete option

3. Admin clicks Delete and confirms the action	4. System archives the user data
5. System removes the user account	6. System confirms the deletion
Alternative Flow	
1-A. Admin tries to delete their own account	2-A. System prevents the action and shows an error message

UC-08 Manage Doctors

Use Case ID	UC-08
Name	Manage Doctors
Actors	Admin
Priority	High
Trigger	Admin navigates to the Manage Doctors section
Summary	Allows the admin to add, edit, or remove doctor profiles in the system
Pre-Conditions	Admin must be logged in
Post-Conditions	Doctor profile is added, updated, or removed from the system
Special Requirements	Each doctor must have a unique ID and assigned specialization
Basic Flow	
Actor Action	System Response
1. Admin opens the Manage Doctors section	2. System displays the list of all doctors
3. Admin adds a new doctor or selects an existing one to edit or delete	4. System displays the relevant form
5. Admin fills in or updates the required information	6. System saves the changes and confirms
Alternative Flow	
1-A. Admin enters incomplete doctor information	2-A. System highlights the missing fields and asks for completion

UC-09 Manage Departments

Use Case ID	UC-09
Name	Manage Departments
Actors	Admin
Priority	Medium
Trigger	Admin navigates to the Manage Departments section
Summary	Allows the admin to create, update, or remove hospital departments
Pre-Conditions	Admin must be logged in
Post-Conditions	Department is created, updated, or removed from the system
Special Requirements	Each department must have a unique name
Basic Flow	
Actor Action	System Response
1. Admin opens the Manage Departments section	2. System displays the list of existing departments
3. Admin adds a new department or edits an existing one	4. System displays the relevant form
5. Admin enters or updates department details	6. System saves the department and confirms
Alternative Flow	
1-A. Admin enters a department name that already exists	2-A. System shows a duplicate name error

UC-10 Define Staffing Levels

Use Case ID	UC-10
Name	Define Staffing Levels
Actors	Admin
Priority	Medium
Trigger	Admin navigates to the Staffing Levels section

Summary	Allows the admin to set the required number of staff for each department
Pre-Conditions	Admin must be logged in. Departments must already be created
Post-Conditions	Staffing levels are saved and visible to department managers
Special Requirements	Staffing levels should be reviewed and updated monthly
Basic Flow	
Actor Action	System Response
1. Admin opens the Staffing Levels section	2. System displays departments with current staffing levels
3. Admin sets or updates the required staff count for each department	4. System validates the entered values
5. Admin saves the staffing levels	6. System updates and confirms the changes
Alternative Flow	
1-A. Admin enters a negative or zero value	2-A. System shows a validation error and asks for a valid number

UC-11 View Staff List

Use Case ID	UC-11
Name	View Staff List
Actors	Admin
Priority	Low
Trigger	Admin clicks View Staff in the administration panel
Summary	Allows the admin to view a complete list of all staff members in the system
Pre-Conditions	Admin must be logged in
Post-Conditions	System displays the full list of staff with their roles and departments

Special Requirements	The list must be searchable and sortable by name, role, and department
Basic Flow	
Actor Action	System Response
1. Admin opens the View Staff section	2. System retrieves and displays the staff list
3. Admin searches or filters the list as needed	4. System updates the display based on the filter
Alternative Flow	
1-A. No staff records found	2-A. System shows a message that no staff records are available

RIPHAH INTERNATIONAL UNIVERSITY

5.4 Patient Access Services

Use Case Diagram

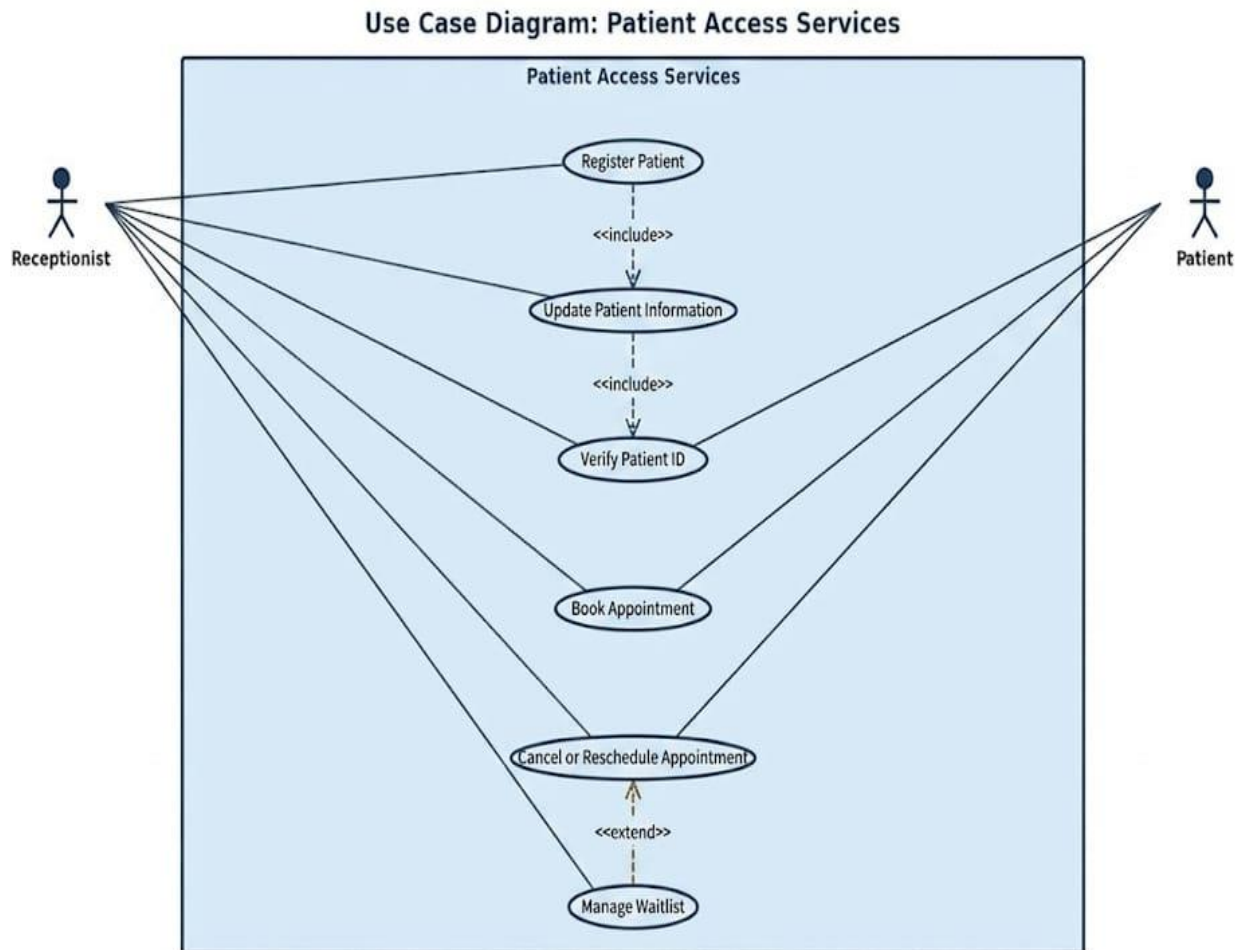


Figure 10: Patient Access Services Use Case Diagram

Activity Diagram

Activity Diagram: Patient Access Services

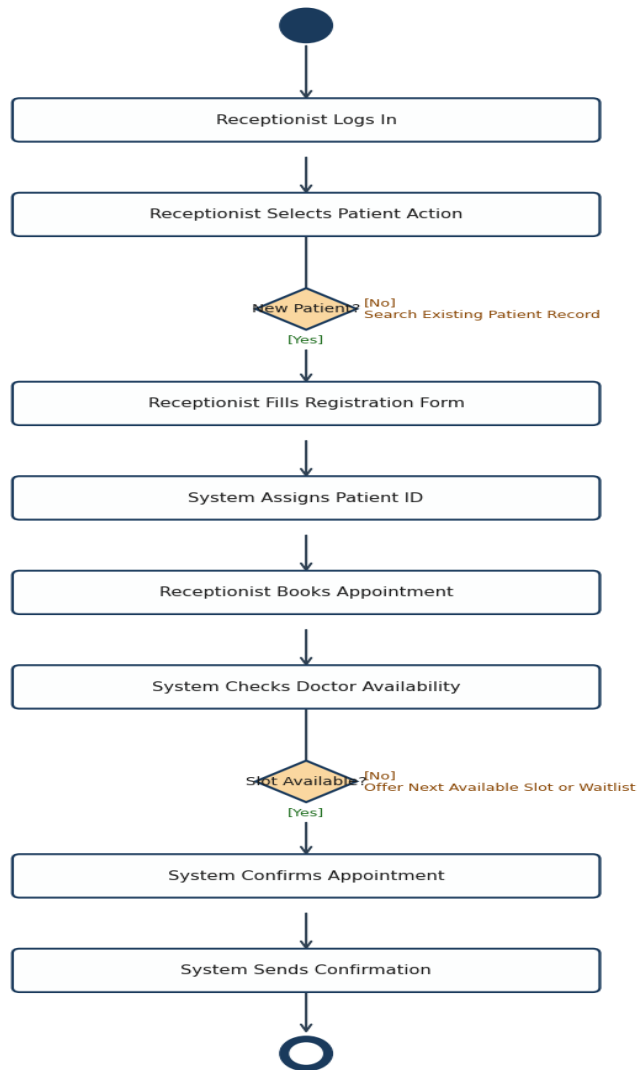


Figure 11: Patient Access Services Activity Diagram

Use Case Description Tables

UC-12 Register Patient

Use Case ID	UC-12
Name	Register Patient
Actors	Receptionist
Priority	High

Trigger	Receptionist clicks Register New Patient
Summary	Allows the receptionist to create a new patient profile in the system
Pre-Conditions	Receptionist must be logged in
Post-Conditions	A new patient profile is created and a unique patient ID is assigned
Special Requirements	Each patient must have a unique ID. Duplicate records must be prevented
Basic Flow	
Actor Action	System Response
1. Receptionist clicks Register New Patient	2. System displays the patient registration form
3. Receptionist enters patient personal and medical details	4. System validates the entered information
5. Receptionist submits the form	6. System creates the patient profile and assigns a unique patient ID
Alternative Flow	
1-A. A patient with the same name and date of birth already exists	2-A. System warns the receptionist of a possible duplicate and asks for confirmation

UC-13 Update Patient Information

Use Case ID	UC-13
Name	Update Patient Information
Actors	Receptionist
Priority	Medium
Trigger	Receptionist searches for a patient and clicks Edit
Summary	Allows the receptionist to update the personal or medical details of an existing patient
Pre-Conditions	Receptionist must be logged in. Patient must already be registered
Post-Conditions	Patient information is updated and saved in the system

Special Requirements	All changes must be logged with the date and time of update
Basic Flow	
Actor Action	System Response
1. Receptionist searches for the patient by name or ID	2. System retrieves and displays the patient profile
3. Receptionist updates the required fields	4. System validates the updated data
5. Receptionist saves the changes	6. System saves the updated information and confirms
Alternative Flow	
1-A. Patient not found in the system	2-A. System shows a not found message and suggests registering the patient

UC-14 Verify Patient ID

Use Case ID	UC-14
Name	Verify Patient ID
Actors	System
Priority	High
Trigger	Any action requires patient identification
Summary	System automatically verifies the patient ID before any patient-related action
Pre-Conditions	A patient ID must be entered or scanned
Post-Conditions	System confirms whether the patient ID is valid or not
Special Requirements	Patient ID verification must complete in under 2 seconds
Basic Flow	
Actor Action	System Response
1. System receives the patient ID	2. System searches the database for a matching record

3. System checks the patient status	4. System confirms the patient identity and returns the result
Alternative Flow	
1-A. Patient ID not found	2-A. System returns an invalid ID message and asks for re-entry

UC-15 Book Appointment

Use Case ID	UC-15
Name	Book Appointment
Actors	Receptionist and Patient
Priority	High
Trigger	Receptionist or Patient clicks Book Appointment
Summary	Allows a receptionist or patient to schedule an appointment with a doctor
Pre-Conditions	Patient must be registered. Doctor must be available at the selected time
Post-Conditions	Appointment is confirmed and added to the doctor schedule
Special Requirements	System must check for scheduling conflicts before confirming the appointment
Basic Flow	
Actor Action	System Response
1. Receptionist or Patient selects a doctor and preferred date and time	2. System checks the doctor availability for the selected slot
3. System confirms the slot is available	4. System books the appointment and assigns a token number
5. Receptionist or Patient confirms the booking	6. System saves the appointment and sends a confirmation
Alternative Flow	
1-A. Selected time slot is not available	2-A. System shows a conflict message and suggests the next available slot

UC-16 Cancel or Reschedule Appointment

Use Case ID	UC-16
Name	Cancel or Reschedule Appointment
Actors	Receptionist and Patient
Priority	Medium
Trigger	Receptionist or Patient selects an appointment and clicks Cancel or Reschedule
Summary	Allows a receptionist or patient to cancel or move an existing appointment to a different time
Pre-Conditions	Appointment must already be booked
Post-Conditions	Appointment is cancelled or rescheduled and the doctor schedule is updated
Special Requirements	Cancellations within 1 hour of appointment time must require admin approval
Basic Flow	
Actor Action	System Response
1. Receptionist or Patient opens the appointment	2. System displays appointment details with Cancel and Reschedule options
3. User selects Cancel or Reschedule and provides a reason	4. System updates the appointment status
5. System notifies the doctor of the change	6. System confirms the cancellation or new schedule to the patient
Alternative Flow	
1-A. User tries to cancel within 1 hour of appointment	2-A. System requests admin approval before proceeding

UC-17 Manage Waitlist

Use Case ID	UC-17
Name	Manage Waitlist
Actors	Receptionist
Priority	Medium

Trigger	No available appointment slots exist and patient still wants to book
Summary	Allows the receptionist to add a patient to a waitlist when no slots are available
Pre-Conditions	Receptionist must be logged in. All doctor slots for the requested time must be full
Post-Conditions	Patient is added to the waitlist and notified when a slot becomes available
Special Requirements	Waitlist is managed on a first-come first-served basis
Basic Flow	
Actor Action	System Response
1. Receptionist confirms all slots are full	2. System displays the option to add the patient to the waitlist
3. Receptionist adds the patient to the waitlist	4. System records the waitlist entry with timestamp
5. A slot becomes available	6. System notifies the receptionist and the patient
Alternative Flow	
1-A. Patient declines to be added to the waitlist	2-A. System records no action and closes the process

5.5 Ward and Bed Management

Use Case Diagram

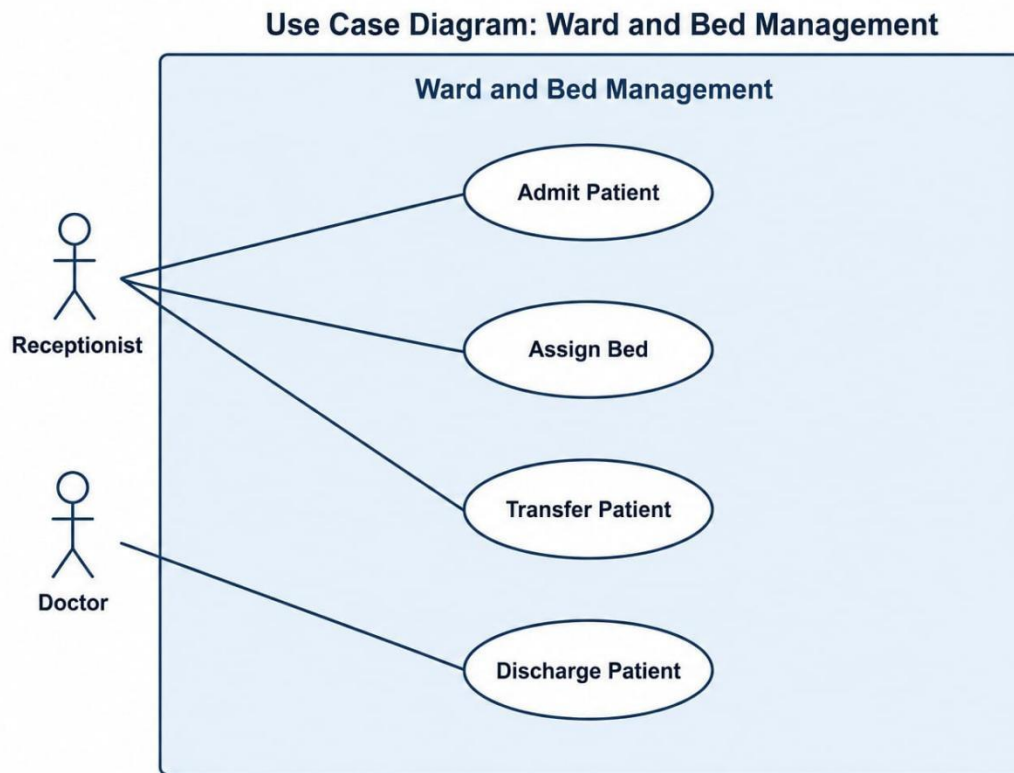


Figure 12: Ward and Bed Management Use Case Diagram

Activity Diagram

Activity Diagram: Ward and Bed Management

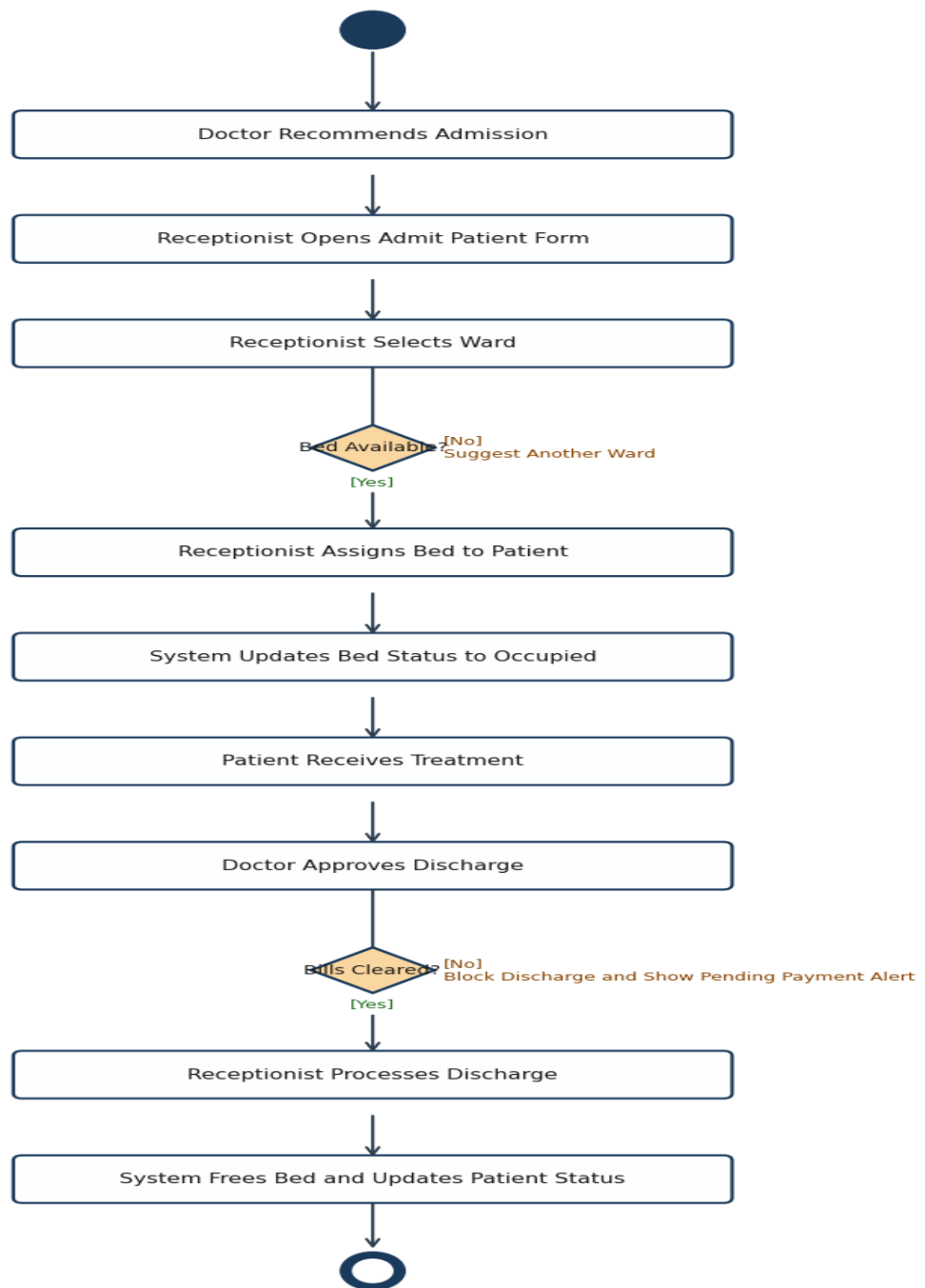


Figure 13: Ward and Bed Management Activity Diagram

Use Case Description Tables

UC-18 Admit Patient

Use Case ID	UC-18
Name	Admit Patient
Actors	Receptionist
Priority	High
Trigger	Doctor recommends patient admission after consultation
Summary	Allows the receptionist to officially admit a patient into the hospital
Pre-Conditions	Patient must be registered. A bed must be available in the required ward
Post-Conditions	Patient is admitted and their status is updated to In-Patient
Special Requirements	Admission details must be linked to the patient medical record
Basic Flow	
Actor Action	System Response
1. Receptionist opens the patient profile	2. System displays the patient details with an Admit option
3. Receptionist selects the required ward and clicks Admit	4. System checks bed availability in the selected ward
5. Receptionist confirms the admission	6. System updates the patient status to In-Patient and assigns a bed
Alternative Flow	
1-A. No beds are available in the selected ward	2-A. System shows a no beds available message and suggests another ward

UC-19 Assign Bed

Use Case ID	UC-19
Name	Assign Bed
Actors	Receptionist

Priority	High
Trigger	Patient is admitted and requires a bed assignment
Summary	Allows the receptionist to assign a specific bed to an admitted patient
Pre-Conditions	Patient must be admitted. A bed must be available in the ward
Post-Conditions	Bed is assigned to the patient and marked as occupied in the system
Special Requirements	Bed assignment must update in real time across all staff views
Basic Flow	
Actor Action	System Response
1. Receptionist opens the ward bed management section	2. System displays available beds in the ward
3. Receptionist selects an available bed and assigns it to the patient	4. System marks the bed as occupied
5. Receptionist confirms the assignment	6. System updates the bed status and links it to the patient record
Alternative Flow	
1-A. All beds in the ward are occupied	2-A. System shows a no available beds message

UC-20 Transfer Patient

Use Case ID	UC-20
Name	Transfer Patient
Actors	Receptionist
Priority	Medium
Trigger	Doctor requests patient transfer to another ward
Summary	Allows the receptionist to move a patient from one ward or bed to another
Pre-Conditions	Patient must be admitted. The destination ward must have an available bed

Post-Conditions	Patient is transferred and the old bed is freed and the new bed is assigned
Special Requirements	Transfer must be logged with date, time, and reason
Basic Flow	
Actor Action	System Response
1. Receptionist opens the patient profile	2. System displays current ward and bed details
3. Receptionist selects the new ward and available bed	4. System checks availability in the destination ward
5. Receptionist confirms the transfer	6. System updates the patient bed assignment and logs the transfer
Alternative Flow	
1-A. No beds available in the destination ward	2-A. System shows a message and suggests waiting or choosing another ward

UC-21 Discharge Patient

Use Case ID	UC-21
Name	Discharge Patient
Actors	Receptionist and Doctor
Priority	High
Trigger	Doctor confirms patient is ready to be discharged
Summary	Allows the doctor or receptionist to discharge a patient and free up the assigned bed
Pre-Conditions	Patient must be admitted. All bills must be cleared before discharge
Post-Conditions	Patient status is updated to Discharged and the bed is freed
Special Requirements	A discharge summary must be generated and attached to the patient record
Basic Flow	
Actor Action	System Response

1. Doctor confirms patient is ready for discharge	2. System checks if all bills are cleared
3. Receptionist processes the discharge	4. System generates a discharge summary
5. Receptionist confirms the discharge	6. System updates patient status to Discharged and frees the bed
Alternative Flow	
1-A. Patient has pending bills	2-A. System blocks discharge and shows a pending payment alert

RIPHAH INTERNATIONAL UNIVERSITY

5.6 Clinical Care

Use Case Diagram

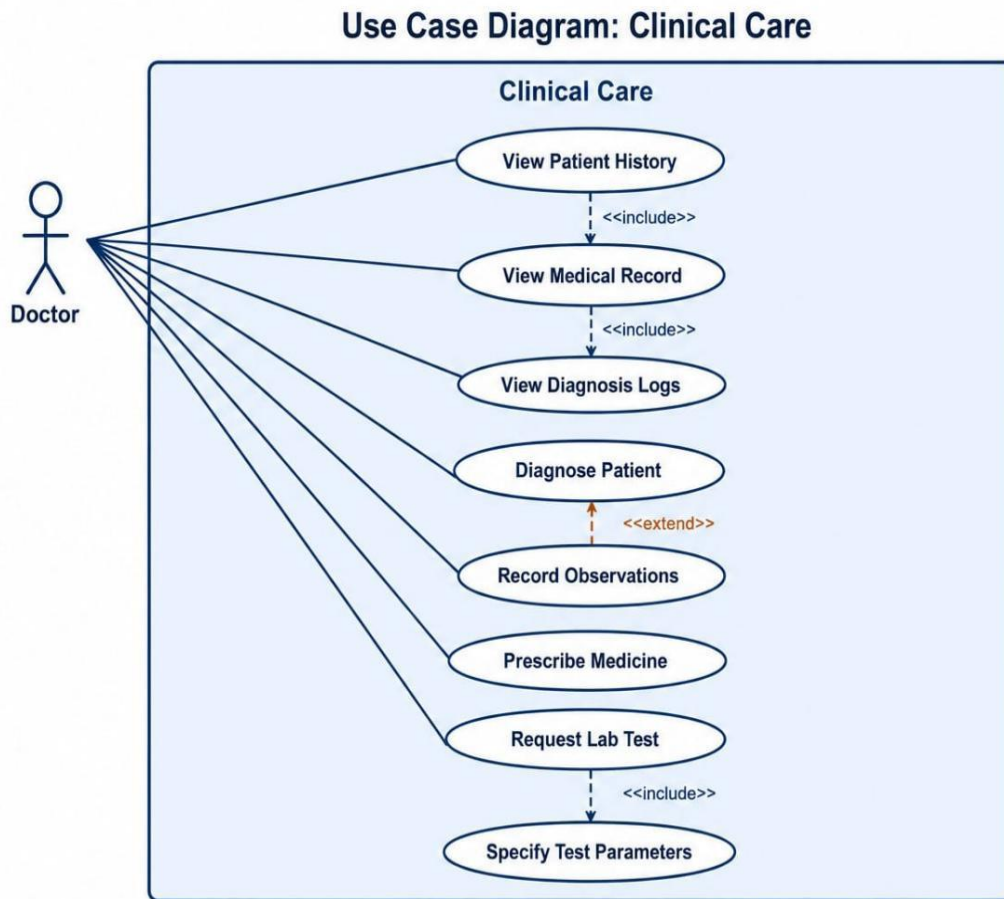


Figure 14: Clinical Care Use Case Diagram

Activity Diagram

Activity Diagram: Clinical Care

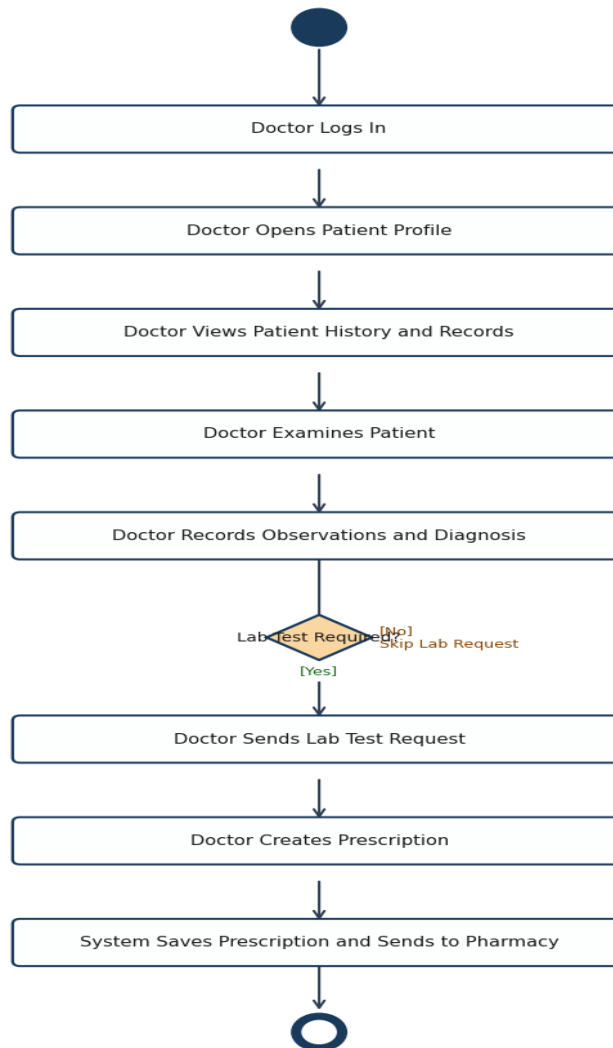


Figure 15: Clinical Care Activity Diagram

Use Case Description Tables

UC-22 View Patient History

Use Case ID	UC-22
Name	View Patient History
Actors	Doctor
Priority	High

Trigger	Doctor opens a patient profile before or during consultation
Summary	Allows the doctor to view the complete medical history of a patient
Pre-Conditions	Doctor must be logged in. Patient must be registered in the system
Post-Conditions	Doctor can view all past visits, diagnoses, prescriptions, and lab results
Special Requirements	Patient history must be read-only for viewing purposes
Basic Flow	
Actor Action	System Response
1. Doctor searches for the patient by name or ID	2. System retrieves and displays the patient profile
3. Doctor opens the medical history section	4. System displays all past visits, diagnoses, and prescriptions
Alternative Flow	
1-A. Patient has no previous records	2-A. System shows a no history available message

UC-23 View Medical Record

Use Case ID	UC-23
Name	View Medical Record
Actors	Doctor
Priority	High
Trigger	Doctor selects a specific visit or record from patient history
Summary	Allows the doctor to view the detailed medical record of a specific patient visit
Pre-Conditions	Doctor must be logged in. Patient medical records must exist
Post-Conditions	Doctor can view full details of the selected medical record

Special Requirements	Records must display in chronological order with the most recent first
Basic Flow	
Actor Action	System Response
1. Doctor opens the patient profile and selects a record	2. System retrieves and displays the full medical record
3. Doctor reviews the record details	4. System allows the doctor to print or download the record if needed
Alternative Flow	
1-A. Selected record is not available	2-A. System shows a record not found message

UC-24 View Diagnosis Logs

Use Case ID	UC-24
Name	View Diagnosis Logs
Actors	Doctor
Priority	Medium
Trigger	Doctor opens the diagnosis history of a patient
Summary	Allows the doctor to view all previous diagnoses made for a patient
Pre-Conditions	Doctor must be logged in. Patient must have at least one diagnosis recorded
Post-Conditions	Doctor can view a complete list of all diagnoses with dates and details
Special Requirements	Diagnosis logs must be read-only and cannot be deleted
Basic Flow	
Actor Action	System Response
1. Doctor opens the patient profile and navigates to Diagnosis Logs	2. System retrieves and displays all diagnosis entries
3. Doctor reviews the diagnosis history	4. System displays each diagnosis with date, doctor name, and details

Alternative Flow	
1-A. No diagnosis records exist for the patient	2-A. System shows a no diagnosis records found message

UC-25 Diagnose Patient

Use Case ID	UC-25
Name	Diagnose Patient
Actors	Doctor
Priority	High
Trigger	Doctor completes examination and clicks Add Diagnosis
Summary	Allows the doctor to record a new diagnosis for the patient after examination
Pre-Conditions	Doctor must be logged in. Patient must be registered and present for consultation
Post-Conditions	Diagnosis is saved and added to the patient medical record
Special Requirements	Diagnosis must include condition name, severity, and notes

Basic Flow	
Actor Action	System Response
1. Doctor examines the patient and clicks Add Diagnosis	2. System displays the diagnosis entry form
3. Doctor enters the diagnosis details including condition and notes	4. System validates the entered information
5. Doctor submits the diagnosis	6. System saves the diagnosis and links it to the patient record

Alternative Flow	
1-A. Doctor submits an incomplete diagnosis form	2-A. System highlights required fields and asks for completion

UC-26 Record Observations

Use Case ID	UC-26
Name	Record Observations
Actors	Doctor
Priority	Medium
Trigger	Doctor clicks Add Observations during or after patient examination
Summary	Allows the doctor to record clinical observations such as vitals and symptoms
Pre-Conditions	Doctor must be logged in. Patient must be registered
Post-Conditions	Observations are saved and attached to the patient record
Special Requirements	Observations must include date, time, and the recording doctor name
Basic Flow	
Actor Action	System Response
1. Doctor opens the patient profile and clicks Add Observations	2. System displays the observations entry form
3. Doctor enters vitals and other clinical observations	4. System validates the entered data
5. Doctor saves the observations	6. System stores the observations and links them to the patient record
Alternative Flow	
1-A. Doctor enters invalid vital values	2-A. System shows a validation error for the specific field

UC-27 Prescribe Medicine

Use Case ID	UC-27
Name	Prescribe Medicine
Actors	Doctor
Priority	High
Trigger	Doctor clicks Add Prescription after diagnosis

Summary	Allows the doctor to create a digital prescription for the patient
Pre-Conditions	Doctor must be logged in. Diagnosis must be recorded for the current visit
Post-Conditions	Prescription is saved and made available to the pharmacy staff
Special Requirements	Prescription must include medicine name, dosage, frequency, and duration
Basic Flow	
Actor Action	System Response
1. Doctor clicks Add Prescription	2. System displays the prescription form with available medicines
3. Doctor selects medicines and enters dosage and duration	4. System validates the prescription details
5. Doctor submits the prescription	6. System saves the prescription and sends it to the pharmacy
Alternative Flow	
1-A. Prescribed medicine is out of stock	2-A. System alerts the doctor and suggests an available alternative

UC-28 Request Lab Test

Use Case ID	UC-28
Name	Request Lab Test
Actors	Doctor
Priority	High
Trigger	Doctor clicks Request Lab Test during consultation
Summary	Allows the doctor to send a lab test request for a patient to the laboratory
Pre-Conditions	Doctor must be logged in. Patient must be registered
Post-Conditions	Lab test request is sent to the laboratory and the lab technician is notified

Special Requirements	Lab test request must include test type, urgency level, and clinical notes
Basic Flow	
Actor Action	System Response
1. Doctor clicks Request Lab Test	2. System displays the lab test request form
3. Doctor selects the required test and enters clinical notes	4. System validates the request
5. Doctor submits the request	6. System sends the request to the laboratory and notifies the lab technician
Alternative Flow	
1-A. Requested test type is not available in the system	2-A. System shows a test not available message and suggests alternatives

UC-29 Specify Test Parameters

Use Case ID	UC-29
Name	Specify Test Parameters
Actors	Doctor
Priority	Medium
Trigger	Doctor adds additional parameters to a lab test request
Summary	Allows the doctor to specify detailed parameters for a requested lab test
Pre-Conditions	Lab test request must already be created
Post-Conditions	Test parameters are saved and attached to the lab request
Special Requirements	Parameters must follow standard medical test guidelines
Basic Flow	
Actor Action	System Response
1. Doctor opens the lab test request	2. System displays the test details with an option to add parameters

3. Doctor enters specific test parameters	4. System validates the parameters
5. Doctor saves the parameters	6. System updates the lab request with the new parameters
Alternative Flow	
1-A. Doctor enters invalid parameter values	2-A. System shows a validation error and asks for correction

RIPHAH INTERNATIONAL UNIVERSITY

5.7 Laboratory Services

Use Case Diagram

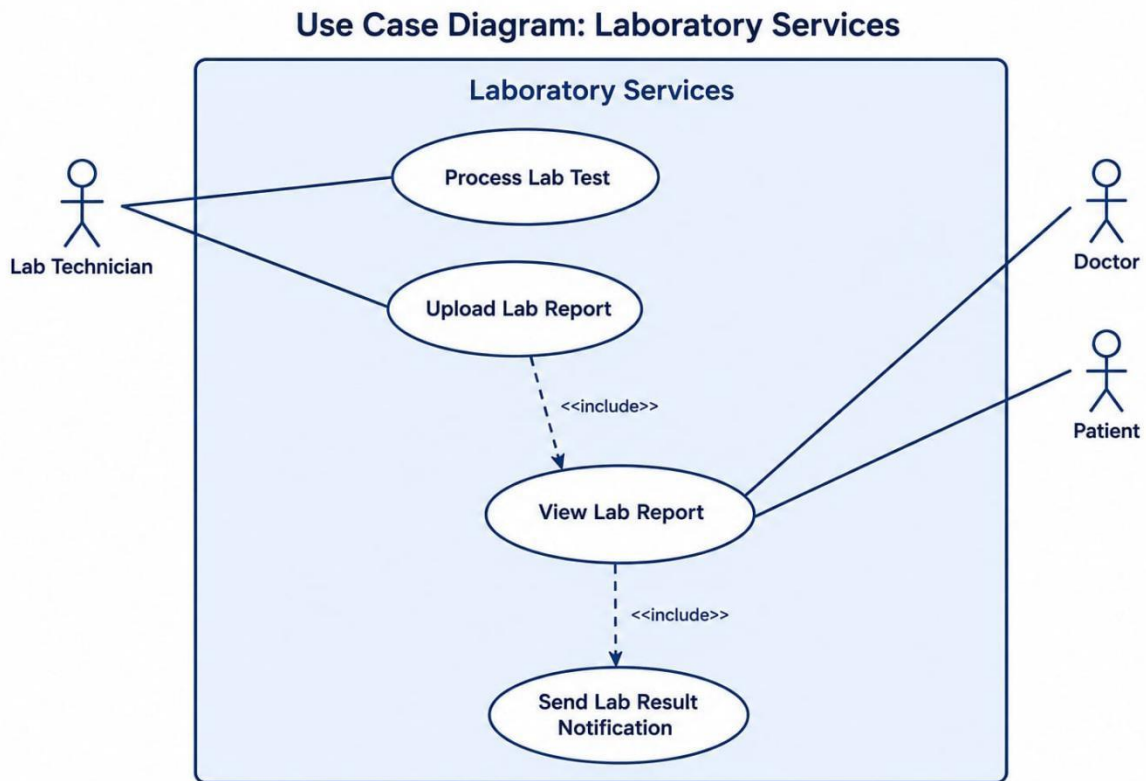


Figure 16: Laboratory Services Use Case Diagram

Activity Diagram

Activity Diagram: Laboratory Services

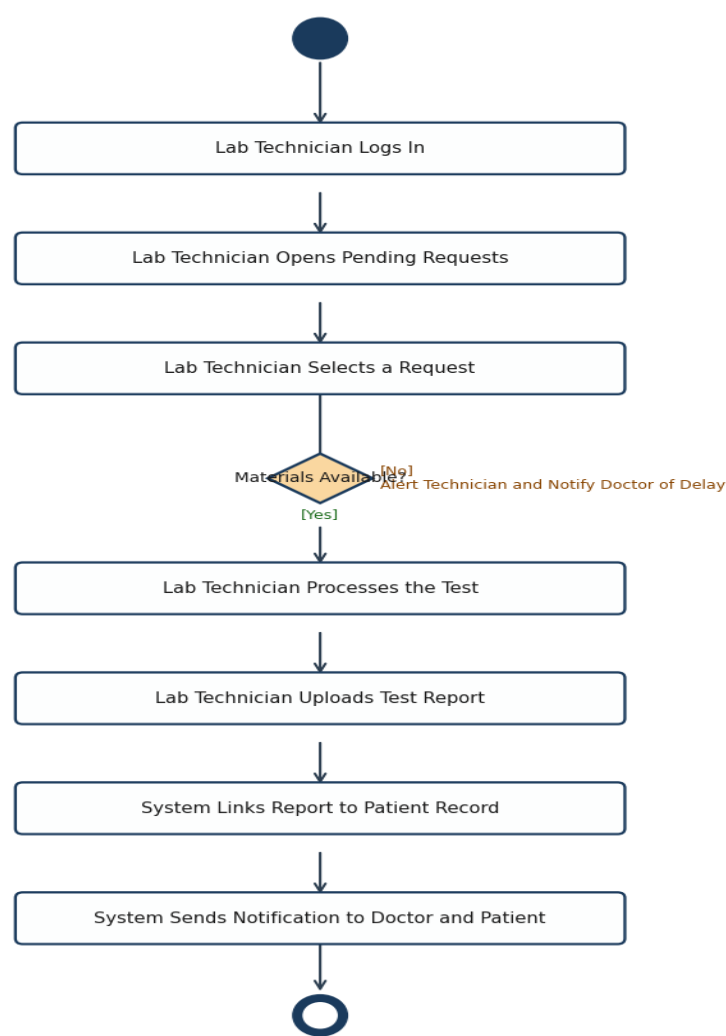


Figure 17: Laboratory Services Activity Diagram

Use Case Description Tables

UC-30 Process Lab Test

Use Case ID	UC-30
Name	Process Lab Test
Actors	Lab Technician
Priority	High

Trigger	Lab Technician receives a lab test request
Summary	Allows the lab technician to process a lab test request sent by a doctor
Pre-Conditions	Lab test request must exist. Required equipment and materials must be available
Post-Conditions	Lab test is processed and results are ready for upload
Special Requirements	Processing time must be recorded for each test
Basic Flow	
Actor Action	System Response
1. Lab Technician opens the pending lab requests	2. System displays all pending test requests
3. Lab Technician selects a request and starts processing	4. System updates the test status to In Progress
5. Lab Technician completes the test	6. System marks the test as Completed and ready for result upload
Alternative Flow	
1-A. Required materials for the test are not available	2-A. System alerts the lab technician and notifies the requesting doctor of the delay

UC-31 Upload Lab Report

Use Case ID	UC-31
Name	Upload Lab Report
Actors	Lab Technician
Priority	High
Trigger	Lab Technician clicks Upload Report after completing a test
Summary	Allows the lab technician to upload the lab test results into the system
Pre-Conditions	Lab test must be processed and results must be ready
Post-Conditions	Lab report is uploaded and linked to the patient record. Doctor is notified

Special Requirements	Lab report must be uploaded in PDF or image format
Basic Flow	
Actor Action	System Response
1. Lab Technician opens the completed test	2. System displays the report upload option
3. Lab Technician uploads the test report	4. System validates the file format and size
5. Lab Technician submits the report	6. System saves the report and links it to the patient record
Alternative Flow	
1-A. File format is not supported	2-A. System shows an invalid file format message and asks for a valid file

UC-32 View Lab Report

Use Case ID	UC-32
Name	View Lab Report
Actors	Doctor and Patient
Priority	High
Trigger	Doctor or Patient opens the lab reports section
Summary	Allows the doctor or patient to view uploaded lab test reports
Pre-Conditions	Lab report must be uploaded. User must have access rights to view the report
Post-Conditions	Lab report is displayed to the authorized user
Special Requirements	Patients can only view their own reports. Doctors can view reports of their patients
Basic Flow	
Actor Action	System Response
1. Doctor or Patient navigates to the lab reports section	2. System retrieves and displays available reports
3. User selects a report to view	4. System opens and displays the full report

Alternative Flow	
1-A. Report is not yet uploaded	2-A. System shows a report not yet available message

UC-33 Send Lab Result Notification

Use Case ID	UC-33
Name	Send Lab Result Notification
Actors	System
Priority	Medium
Trigger	Lab report is uploaded into the system
Summary	System automatically notifies the doctor and patient when a lab report is ready
Pre-Conditions	Lab report must be uploaded
Post-Conditions	Doctor and patient receive a notification that the lab report is available
Special Requirements	Notifications must be sent within 1 minute of report upload

Basic Flow	
Actor Action	System Response
1. Lab report is uploaded by lab technician	2. System detects the new report upload
3. System identifies the requesting doctor and the patient	4. System sends a notification to both the doctor and the patient

Alternative Flow	
1-A. Doctor or patient notification fails to deliver	2-A. System retries the notification up to 3 times

5.8 Pharmacy Services

Use Case Diagram

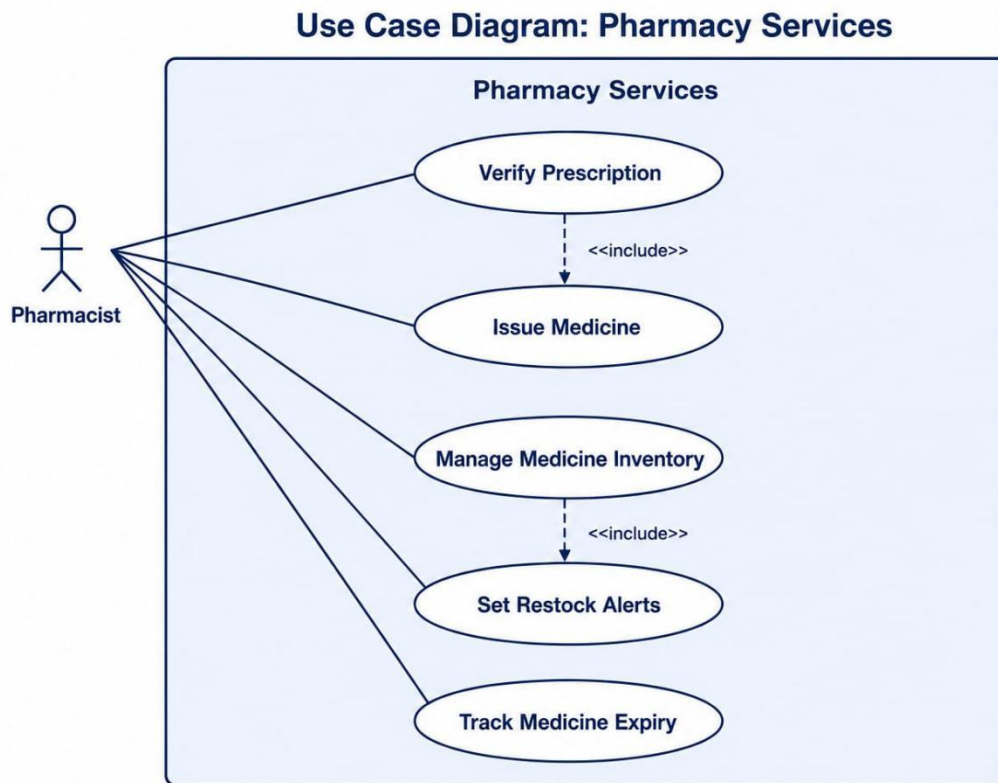


Figure 18: Pharmacy Services Use Case Diagram

Activity Diagram

Activity Diagram: Pharmacy Services

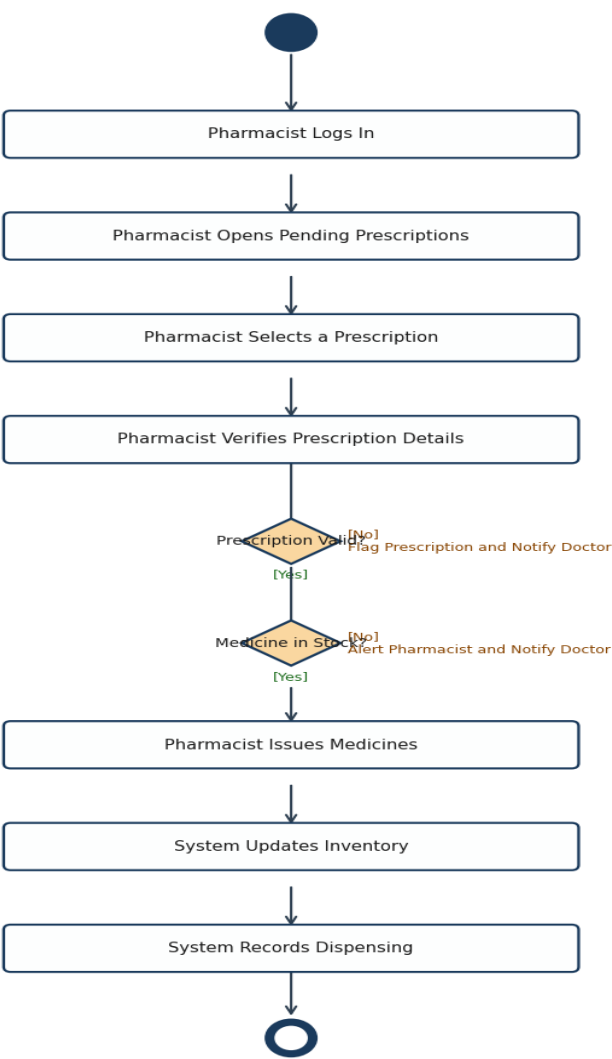


Figure 19: Pharmacy Services Activity Diagram

Use Case Description Tables

UC-34 Verify Prescription

Use Case ID	UC-34
Name	Verify Prescription
Actors	Pharmacist
Priority	High

Trigger	Pharmacist receives a prescription from the system
Summary	Allows the pharmacist to verify a digital prescription before dispensing medicines
Pre-Conditions	Prescription must be created by a doctor and sent to the pharmacy
Post-Conditions	Prescription is verified and approved or flagged for review
Special Requirements	Prescription must be verified within 15 minutes of receipt
Basic Flow	
Actor Action	System Response
1. Pharmacist opens the pending prescriptions	2. System displays all received prescriptions
3. Pharmacist selects a prescription and reviews the details	4. System displays medicine name, dosage, and doctor information
5. Pharmacist verifies the prescription	6. System marks the prescription as Verified and ready for dispensing
Alternative Flow	
1-A. Prescription contains an error or unclear instruction	2-A. System allows the pharmacist to flag the prescription and notify the doctor

UC-35 Issue Medicine

Use Case ID	UC-35
Name	Issue Medicine
Actors	Pharmacist
Priority	High
Trigger	Pharmacist clicks Issue Medicine after prescription is verified
Summary	Allows the pharmacist to dispense medicines to the patient based on the prescription
Pre-Conditions	Prescription must be verified. Medicines must be available in stock

Post-Conditions	Medicines are dispensed and stock is updated automatically
Special Requirements	Issued medicines must be recorded with quantity and date of dispensing
Basic Flow	
Actor Action	System Response
1. Pharmacist opens the verified prescription	2. System displays the list of prescribed medicines
3. Pharmacist confirms the medicines to be issued	4. System checks stock availability for each medicine
5. Pharmacist issues the medicines	6. System updates the inventory and records the dispensing
Alternative Flow	
1-A. A prescribed medicine is out of stock	2-A. System shows an out of stock alert and notifies the pharmacist to contact the doctor

UC-36 Manage Medicine Inventory

Use Case ID	UC-36
Name	Manage Medicine Inventory
Actors	Pharmacist
Priority	High
Trigger	Pharmacist navigates to the Medicine Inventory section
Summary	Allows the pharmacist to view, add, update, or remove medicines in the inventory
Pre-Conditions	Pharmacist must be logged in
Post-Conditions	Inventory is updated with the latest medicine stock information
Special Requirements	Inventory must always reflect real time stock levels
Basic Flow	
Actor Action	System Response

1. Pharmacist opens the Medicine Inventory section	2. System displays all medicines with current stock levels
3. Pharmacist adds a new medicine or updates existing stock	4. System validates the entered information
5. Pharmacist saves the changes	6. System updates the inventory and confirms
Alternative Flow	
1-A. Pharmacist enters a negative stock quantity	2-A. System shows a validation error and asks for a valid quantity

UC-37 Set Restock Alerts

Use Case ID	UC-37
Name	Set Restock Alerts
Actors	Pharmacist
Priority	Medium
Trigger	Pharmacist clicks Set Alert on a medicine in the inventory
Summary	Allows the pharmacist to set a minimum stock level that triggers a restock alert
Pre-Conditions	Pharmacist must be logged in. Medicine must exist in the inventory
Post-Conditions	Restock alert is set and the system sends an alert when stock reaches the minimum level
Special Requirements	Alerts must be reviewed and updated whenever stock levels change significantly
Basic Flow	
Actor Action	System Response
1. Pharmacist opens the medicine details	2. System displays the medicine information with a Set Alert option
3. Pharmacist sets the minimum stock level	4. System saves the threshold value
5. Stock drops to or below the threshold	6. System automatically sends a restock alert to the pharmacist

Alternative Flow	
1-A. Pharmacist sets an invalid threshold value	2-A. System shows a validation error and asks for a valid number

UC-38 Track Medicine Expiry

Use Case ID	UC-38
Name	Track Medicine Expiry
Actors	Pharmacist
Priority	Medium
Trigger	Pharmacist opens the Medicine Expiry Tracker section
Summary	Allows the pharmacist to monitor medicine expiry dates and remove expired stock
Pre-Conditions	Pharmacist must be logged in. Medicines with expiry dates must exist in the inventory
Post-Conditions	Expired medicines are flagged and removed from the active inventory
Special Requirements	System must automatically flag medicines expiring within 30 days

Basic Flow	
Actor Action	System Response
1. Pharmacist opens the Medicine Expiry Tracker	2. System displays all medicines with expiry dates highlighted by urgency
3. Pharmacist reviews expiring or expired medicines	4. System shows medicines expiring within 30 days in yellow and expired in red
5. Pharmacist removes expired medicines	6. System updates the inventory and logs the removal

Alternative Flow	
1-A. No medicines are near expiry	2-A. System shows an all medicines are within valid date message

5.9 Financial Services

Use Case Diagram

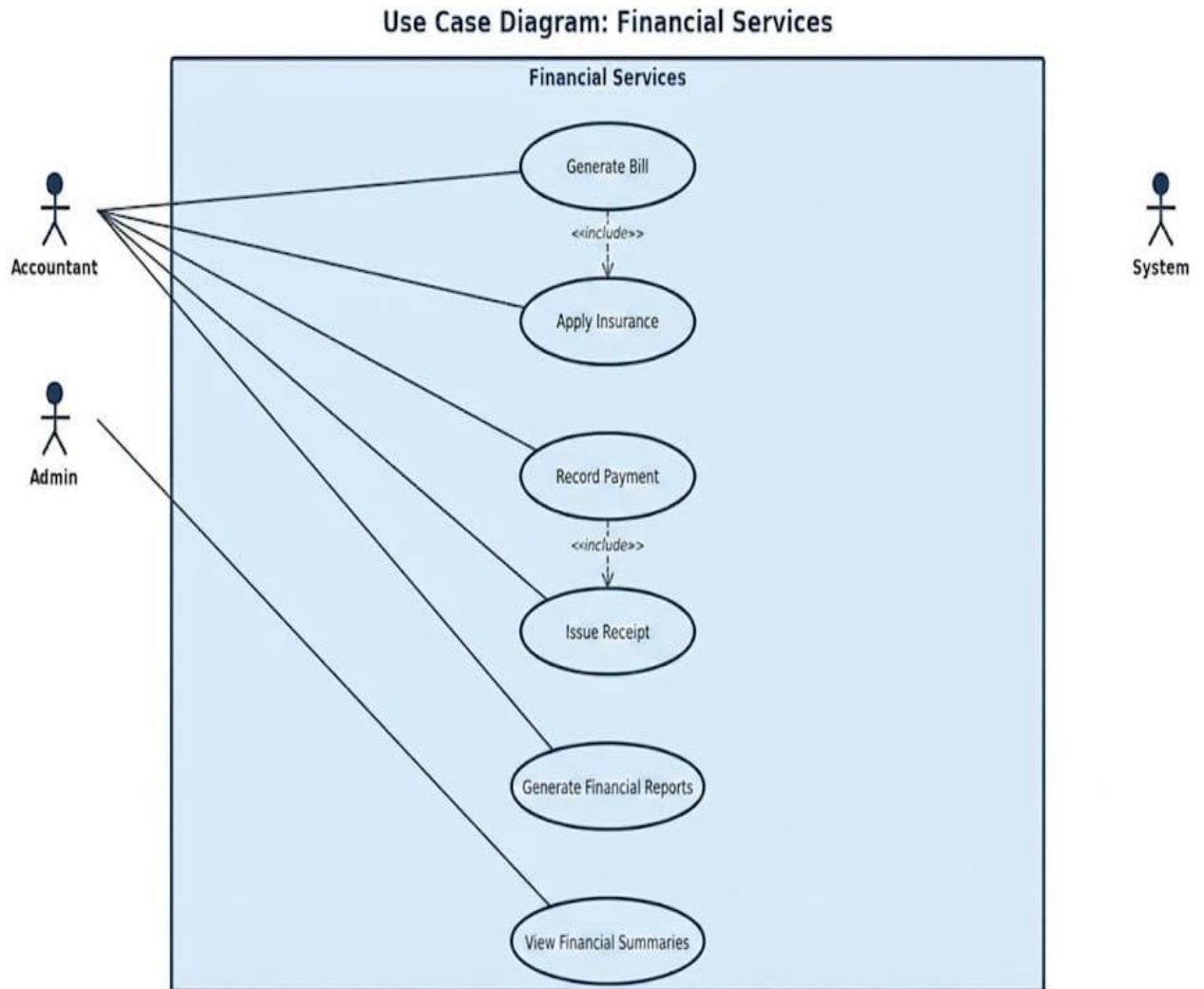


Figure 20: Financial Services Use Case Diagram

Activity Diagram

Activity Diagram: Financial Services

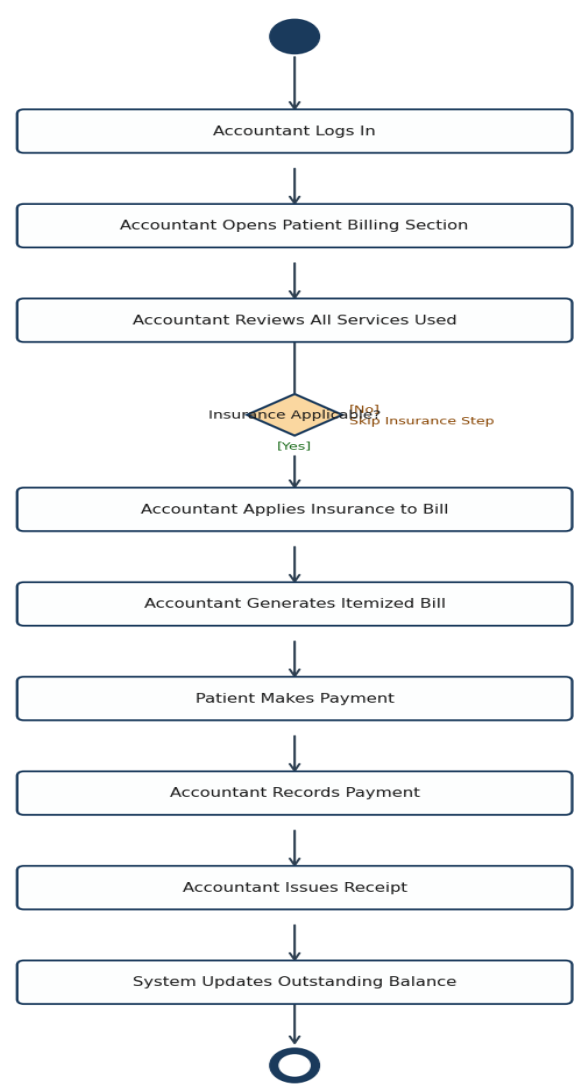


Figure 21: Financial Services Activity Diagram

Use Case Description Tables

UC-39 Generate Bill

Use Case ID	UC-39
Name	Generate Bill
Actors	Accountant
Priority	High

Trigger	Accountant clicks Generate Bill for a patient
Summary	Allows the accountant to generate an itemized bill for all services received by a patient
Pre-Conditions	Accountant must be logged in. Patient services must be recorded in the system
Post-Conditions	An itemized bill is generated and made available to the patient
Special Requirements	Bill must include consultation fees, lab charges, medicine costs, and room charges separately
Basic Flow	
Actor Action	System Response
1. Accountant opens the patient billing section	2. System retrieves all services used by the patient
3. Accountant reviews the service list and clicks Generate Bill	4. System calculates the total amount for all services
5. Accountant confirms the bill	6. System generates the itemized bill and saves it to the patient record
Alternative Flow	
1-A. Some services are not yet recorded	2-A. System shows a warning and asks the accountant to verify before generating

UC-40 Apply Insurance

Use Case ID	UC-40
Name	Apply Insurance
Actors	Accountant
Priority	Medium
Trigger	Patient provides insurance information and accountant clicks Apply Insurance
Summary	Allows the accountant to apply insurance coverage to reduce the patient bill
Pre-Conditions	Bill must be generated. Patient must have valid insurance information

Post-Conditions	Insurance is applied and the remaining payable amount is updated on the bill
Special Requirements	Insurance details must be verified before being applied to the bill
Basic Flow	
Actor Action	System Response
1. Accountant opens the patient bill	2. System displays the total bill amount with an Apply Insurance option
3. Accountant enters the insurance details	4. System verifies and calculates the insurance coverage
5. Accountant confirms the insurance application	6. System updates the bill with the reduced payable amount
Alternative Flow	
1-A. Insurance details are invalid or expired	2-A. System shows an invalid insurance message and the full amount remains payable

UC-41 Record Payment

Use Case ID	UC-41
Name	Record Payment
Actors	Accountant
Priority	High
Trigger	Patient makes a payment and accountant clicks Record Payment
Summary	Allows the accountant to record a payment made by the patient against their bill
Pre-Conditions	Bill must be generated. Payment must be received from the patient
Post-Conditions	Payment is recorded and the remaining balance is updated on the bill
Special Requirements	Partial payments must be allowed and tracked separately
Basic Flow	

Actor Action	System Response
1. Accountant opens the patient bill	2. System displays the outstanding amount
3. Accountant enters the payment amount and method	4. System validates the payment details
5. Accountant confirms the payment	6. System records the payment and updates the outstanding balance
Alternative Flow	
1-A. Payment amount exceeds the outstanding balance	2-A. System shows a payment exceeds due amount error

UC-42 Issue Receipt

Use Case ID	UC-42
Name	Issue Receipt
Actors	Accountant
Priority	High
Trigger	Accountant clicks Issue Receipt after payment is recorded
Summary	Allows the accountant to generate and issue a payment receipt to the patient
Pre-Conditions	Payment must be recorded in the system
Post-Conditions	Receipt is generated and provided to the patient
Special Requirements	Receipt must include payment date, amount paid, and remaining balance if any
Basic Flow	
Actor Action	System Response
1. Accountant opens the payment record	2. System displays the payment details with an Issue Receipt option
3. Accountant clicks Issue Receipt	4. System generates the receipt with all payment details
5. Accountant prints or sends the receipt to the patient	6. System saves a copy of the receipt in the patient record

Alternative Flow	
1-A. No payment has been recorded yet	2-A. System disables the Issue Receipt option and shows a no payment recorded message

UC-43 Generate Financial Reports

Use Case ID	UC-43
Name	Generate Financial Reports
Actors	Accountant
Priority	Medium
Trigger	Accountant clicks Generate Report in the financial reports section
Summary	Allows the accountant to generate daily, weekly, or monthly financial reports
Pre-Conditions	Accountant must be logged in. Financial transactions must exist in the system
Post-Conditions	Financial report is generated and available for download
Special Requirements	Reports must be exportable in PDF format

Basic Flow	
Actor Action	System Response
1. Accountant opens the financial reports section	2. System displays report type and date range options
3. Accountant selects the report type and date range	4. System retrieves the relevant financial data
5. Accountant clicks Generate	6. System generates the report and makes it available for download

Alternative Flow	
1-A. No financial data exists for the selected period	2-A. System shows a no data available for the selected period message

UC-44 View Financial Summaries

Use Case ID	UC-44
-------------	-------

Name	View Financial Summaries
Actors	Accountant and Admin
Priority	Medium
Trigger	Accountant or Admin opens the financial summaries section
Summary	Allows the accountant or admin to view a quick summary of hospital financial performance
Pre-Conditions	User must be logged in with accountant or admin role
Post-Conditions	Financial summary is displayed with total revenue, payments received, and outstanding amounts
Special Requirements	Summary must reflect real-time data and be updated automatically
Basic Flow	
Actor Action	System Response
1. User opens the financial summaries section	2. System retrieves and displays key financial figures
3. User reviews the summary	4. System shows total revenue, payments, and outstanding balances
Alternative Flow	
1-A. No financial data is available	2-A. System shows a no financial records found message

5.10 System and Admin Services

Use Case Diagram

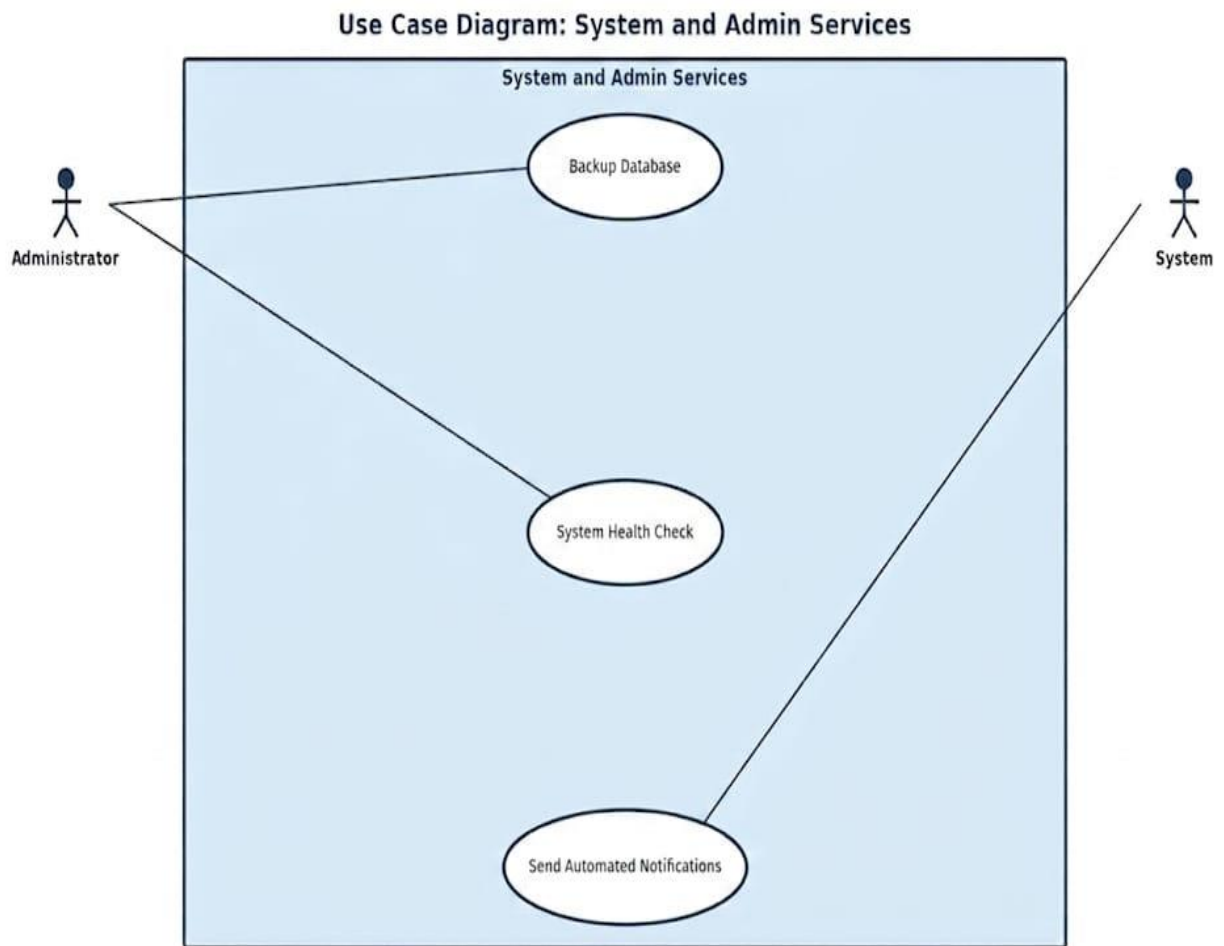


Figure 22: System and Admin Services Use Case Diagram

Activity Diagram

Activity Diagram: System and Admin Services

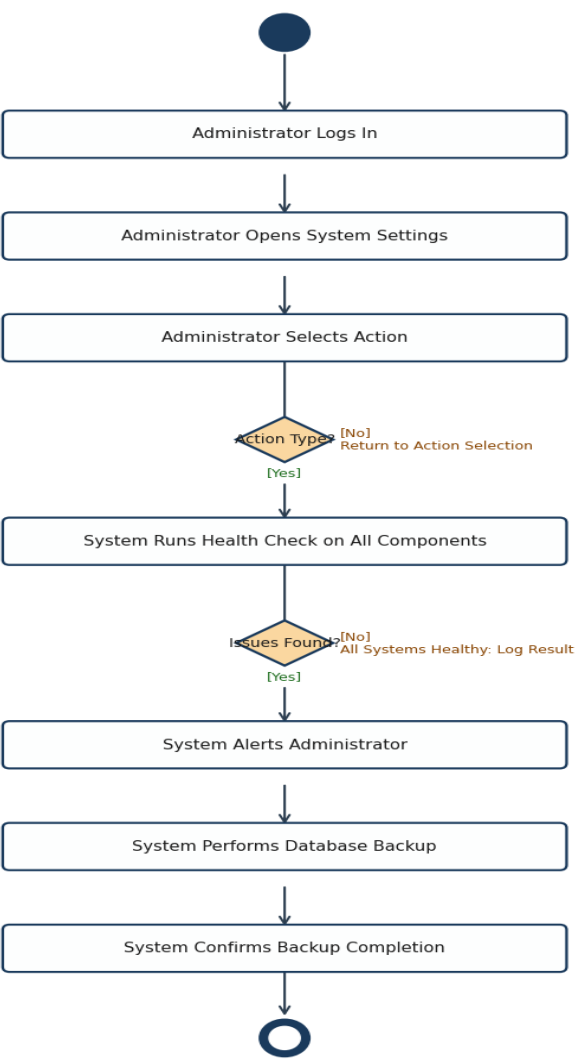


Figure 23: System and Admin Services Activity Diagram

Use Case Description Tables

UC-45 Backup Database

Use Case ID	UC-45
Name	Backup Database
Actors	Administrator
Priority	High

Trigger	Administrator clicks Backup Now or system triggers an automatic backup
Summary	Allows the administrator to create a backup of the entire system database
Pre-Conditions	Administrator must be logged in with system-level access
Post-Conditions	Database backup is created and stored securely
Special Requirements	Automatic backup must run daily at midnight. Manual backup is available anytime
Basic Flow	
Actor Action	System Response
1. Administrator opens the system settings and clicks Backup Now	2. System initiates the backup process
3. System compresses and saves all database records	4. System generates a backup confirmation with timestamp
5. Administrator verifies the backup completion	6. System stores the backup securely and logs the action
Alternative Flow	
1-A. Backup process fails due to storage issues	2-A. System sends an alert to the administrator and logs the failure

UC-46 System Health Check

Use Case ID	UC-46
Name	System Health Check
Actors	Administrator
Priority	Medium
Trigger	Administrator opens the System Health section or system runs an automated check
Summary	Allows the administrator to monitor the overall health and performance of the system
Pre-Conditions	Administrator must be logged in with system-level access

Post-Conditions	System health report is displayed showing status of all components
Special Requirements	Health check must run automatically every hour and alert the administrator if any issue is detected
Basic Flow	
Actor Action	System Response
1. Administrator opens the System Health section	2. System runs a diagnostic check on all components
3. System checks server status, database connectivity, and performance	4. System displays a health report with status indicators
5. Administrator reviews the report	6. System logs the health check result
Alternative Flow	
1-A. A system component is not responding	2-A. System highlights the issue and sends an alert to the administrator

UC-47 Send Automated Notifications

Use Case ID	UC-47
Name	Send Automated Notifications
Actors	System
Priority	Medium
Trigger	A system event occurs that requires notification such as appointment reminder or lab result
Summary	System automatically sends notifications to relevant users when specific events occur
Pre-Conditions	A triggering event must occur such as appointment booking, lab report upload, or low stock
Post-Conditions	Notification is sent to the relevant user and logged in the system
Special Requirements	Notifications must be delivered within 1 minute of the triggering event
Basic Flow	

Actor Action	System Response
1. A triggering event occurs in the system	2. System identifies the relevant users to be notified
3. System prepares the notification message	4. System sends the notification to the identified users
5. Users receive the notification	6. System logs the notification with timestamp
Alternative Flow	
1-A. Notification delivery fails	2-A. System retries up to 3 times and logs the failure if all attempts fail

5.11 Patient Self Services

Use Case Diagram

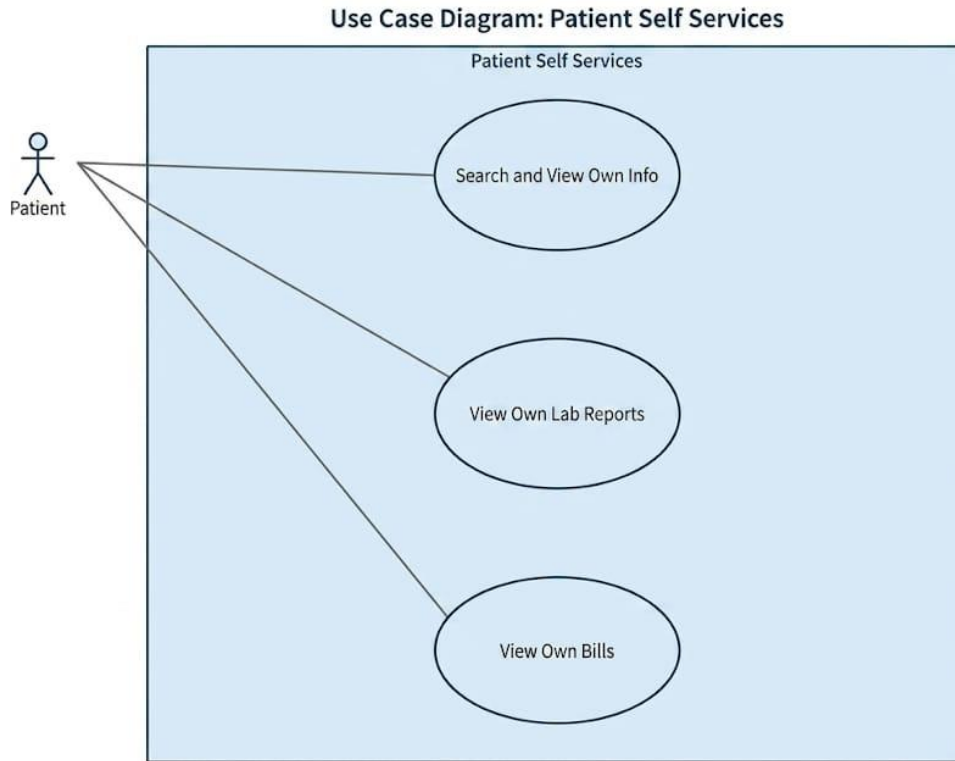


Figure 24: Patient Self Services Use Case Diagram

Activity Diagram

Activity Diagram: Patient Self Services

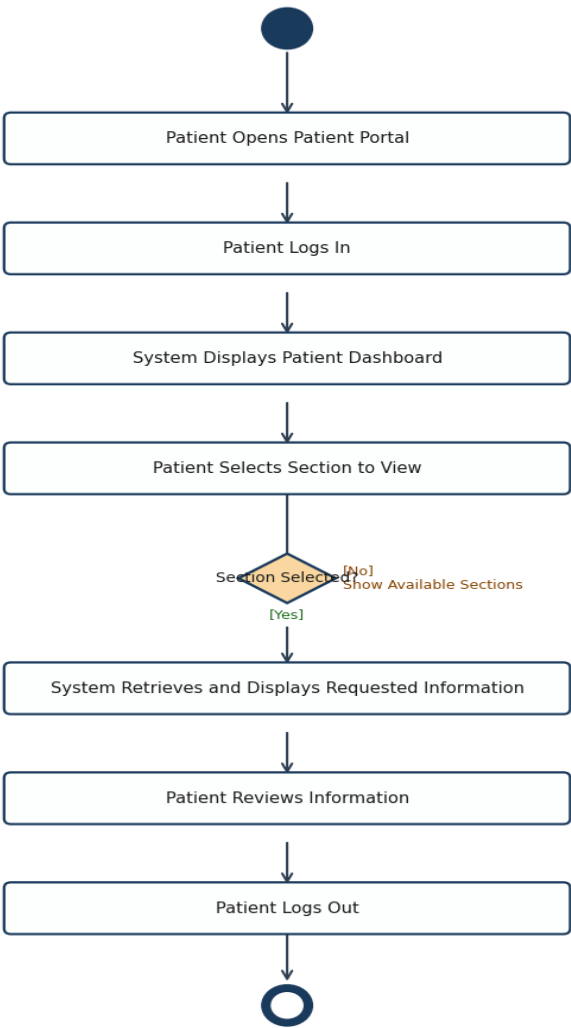


Figure 25: Patient Self Services Activity Diagram

Use Case Description Tables

UC-48 Search and View Own Info

Use Case ID	UC-48
Name	Search and View Own Info
Actors	Patient
Priority	Medium

Trigger	Patient logs into the patient portal and opens their profile
Summary	Allows the patient to search and view their own personal and medical information
Pre-Conditions	Patient must be registered and logged in to the patient portal
Post-Conditions	Patient can view their profile including personal details and medical records
Special Requirements	Patients can only view their own information and cannot edit it
Basic Flow	
Actor Action	System Response
1. Patient logs into the patient portal	2. System displays the patient dashboard
3. Patient opens their profile	4. System retrieves and displays the patient personal and medical information
Alternative Flow	
1-A. Patient profile has no records	2-A. System shows a no records found message

UC-49 View Own Lab Reports

Use Case ID	UC-49
Name	View Own Lab Reports
Actors	Patient
Priority	Medium
Trigger	Patient clicks View Lab Reports in their patient portal
Summary	Allows the patient to view their own lab test reports uploaded by the lab technician
Pre-Conditions	Patient must be logged in. Lab reports must be uploaded and linked to the patient
Post-Conditions	Patient can view their lab reports
Special Requirements	Patients can only view their own reports and cannot modify them

Basic Flow	
Actor Action	System Response
1. Patient opens the Lab Reports section	2. System retrieves and displays all lab reports linked to the patient
3. Patient selects a report to view	4. System opens and displays the full report
Alternative Flow	
1-A. No lab reports are available	2-A. System shows a no lab reports available message

UC-50 View Own Bills

Use Case ID	UC-50
Name	View Own Bills
Actors	Patient
Priority	Medium
Trigger	Patient clicks View Bills in their patient portal
Summary	Allows the patient to view their billing history and outstanding payments
Pre-Conditions	Patient must be logged in. Bills must be generated and linked to the patient
Post-Conditions	Patient can view itemized bills and payment status
Special Requirements	Patients can view but cannot edit or delete their bills
Basic Flow	
Actor Action	System Response
1. Patient opens the Bills section	2. System retrieves and displays all bills linked to the patient
3. Patient selects a bill to view	4. System displays the full itemized bill with payment status
Alternative Flow	
1-A. No bills are available	2-A. System shows a no billing records found message